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Sentiment Analysis and Opinion Mining of FIFA 23 Steam Reviews

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Assessment Form

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1. Introduction

With the surge of online gaming communities and digital distribution platforms, a substantial amount of user-generated content has emerged in the form of reviews, comments, and ratings. Players can post their experience, opinion, and satisfaction about video games on gaming sites like Steam. These reviews can be valuable resources for developers, publishers, and researchers to gain insights into player preferences, strengths and weaknesses of the product, and future games.

However, with how massive amount of user reviews there are, it's hard, time-consuming and inefficient to manually process. As a result, sentiment analysis has become an important technique in the field of Social Media Computing and Natural Language Processing (NLP). Sentiment analysis is the classification of opinions found in textual information, into positive, negative and neutral categories.

So in this project, we will analyse user reviews of FIFA 23 that have been gathered from the Steam platform and are available on the Kaggle dataset “FIFA23 Steam Reviews.” In this study, sentiment analysis methods are used to segment players' opinions and determine the trends of user behaviour. The project also examines the correlation between sentiment and other factors related to the game, including: recent playtime, number of games owned, date of creation of the review, and the recommendation status of the review.

The project involves various steps of text preprocessing, feature engineering, machine learning, and data visualization, all aimed at generating valuable insights from social media data. Together with visualization methods, traditional machine-learning models like Logistic Regression are used to assess player sentiment about FIFA 23.

Furthermore, this research highlights the significance of sentiment analysis in today's digital era where user-generated content has a pivotal role in shaping both consumer choices and public opinion. To understand the consequences of players satisfaction trends in FIFA 23 and to gain insights into the reactions of gaming communities to the FIFA 23 game quality, updates, and gameplay experiences, this project analyses the FIFA 23 reviews.

1.1 Project Overview

The aim of this project is to analyse the sentiment and opinions of players in the FIFA 23 Steam reviews using Natural Language Processing and machine learning techniques. There are 16 columns and 25,737 Steam review records in the raw dataset, such as the reviews text, review date, recommendation status, playtime, number of games owned, and other metadata related to the users.

In this project the primary supervised label is `voted_up`, True denotes a positive Steam recommendation, False denotes a negative Steam recommendation. Finally, following the deletion of the missing/rejected/review texts without content and the elimination of duplicate review texts, there are 20,050 English Review texts. Of these reviews, 9310 are positive and 10740 are negative. The time span of the reviews listed is from 29 September 2022 to 12 March 2023, making it possible to analyze the broad sentiment of the reviews and the sentiment trends over time.

The project builds an end-to-end NLP pipeline from data cleaning and text preprocessing to exploratory data analysis, sentiment analysis, hypothesis testing, extracting features using TF-IDF, traditional machine learning models, transformer based sentiment benchmark, opinion phrase extraction, and lightweight aspect-based sentiment analysis. It also aims to evaluate reviews for their positivity or negativity, as well as analyzing why players are satisfied and dissatisfied, particularly with the gameplay, graphics, bugs, online servers, content, and monetization.

2. Problem Statement

Online gaming platforms have thousands of player reviews a day, which can be very useful to get some insight on the quality of the game, bugs, updates, graphics and overall satisfaction of the player. One of the most popular football simulation games, FIFA 23 is the subject of numerous reviews by players in Steam. Player reviews and experiences are a valuable resource for game developers and researchers that contribute to these reviews.

Analysing a huge number of reviews manually is inefficient and impractical, however, because of the time and effort it would take. Traditional ways of processing large scale textual data effectively are not feasible, particularly when opinions evolve over time. Hence, automated tools and methods for sentiment analysis are required to properly categorize and analyze the feedback from players.

The goal of this study is to use sentiment analysis methods to categorize the FIFA 23 Steam reviews as positive or negative sentiment and to uncover patterns and links between player actions and the sentiment of the reviews. The following hypotheses are explored in this project:

- Hypothesis 1

H1: The more players play the past two weeks, the more likely they are to leave positive reviews for FIFA 23.

The null hypothesis (H0) for this test is that there is no significant correlation between the amount of time played in the past 2 weeks and sentiment towards the FIFA 23 game.

- Hypothesis 2

H2: But their negative rating is more likely to come from players with more purchases, indicating more critical comments with the larger gaming portfolio. If the number of games owned is not related to review sentiment for FIFA 23, then the null hypothesis (H0) is accepted.

- Hypothesis 3

H3: The sentiment of players with respect to FIFA 23 varies with the date of the review.

The null Hypotheses (H0): Player sentiment does not significantly differ over time.

- Hypothesis 4

H4: The results showed that the sentiment in the written reviews has a significant relationship with the Steam voted_up recommendation label for FIFA 23.

The null hypothesis (H0) is that the sentiment in written reviews and the Steam voted_up recommendation labels for FIFA 23 are not significantly related.

Steam has a voted_up label to indicate emotional sentiment, but this does not necessarily indicate emotional sentiment. Players can post a positive review but not recommend the game if there is a price issue, bugs, online or anti-cheat issues. Likewise, reviews can be sarcastic, contain slang, be short or have mixed reactions that are hard to make sense of with a simple sentiment analysis.

As such, this project does not assume that voted_up is a perfect measure of human emotion as a recommendation signal, and only considers it as an observation of a recommendation signal. The goal of this study is to classify the FIFA 23 Steam reviews as positive and negative reviews by applying the label "voted_up" from the Supervised Machine Learning model, compare the results of the Supervised Machine Learning model with the Lexicon-based written sentiment, and determine the top positive and negative game aspects discussed.

Another aspect of the project is to determine if player behaviour attributes (e.g., number of games owned, hours of recent playtime) are significantly correlated with the sentiment of the reviews.

3. Literature Review

One of the most important research fields within Natural Language Processing (NLP) and Social Media Computing is sentiment analysis. It is about how emotions, opinions and attitudes are identified, extracted and classified from textual data. As digital communication platforms like Twitter, Reddit, Steam, YouTube and Facebook expand at an extremely fast pace, there is a huge amount of user-generated content created every day. These platforms offer valuable information on customer satisfaction, public perception, and user experiences when it comes to product and services.

Liu (2012) defines sentiment analysis, or opinion mining, as the “computational analysis of opinions, evaluations, feelings, and attitudes mentioned in texts.” Sentiment analysis is already used across a variety of sectors such as e-commerce, healthcare, politics, entertainment and games. Sentiment analysis techniques are employed by companies and researchers to gain insights into customer behaviour, enhance products, and quickly detect public trends.

In the gaming industry, online reviews can have a significant impact, as they can influence the decisions of players regarding the game they intend to purchase and the reputation of the game. Using modern gaming platforms, players can give instant feedback on the quality of the game, its graphics, bugs, online servers, the balance, and the monetization systems. One of the biggest digital game distribution sites where players can post reviews and recommendations of the games they buy and play is Steam.

Steam reviews are an ideal option for conducting sentiment analysis as they include textual reviews and metadata including whether a product was recommended, how many hours were spent playing, when the review was written, and how many users interacted with the game. The other attributes enable researchers to analyze not only sentiment classification but also player patterns of behaviour. For instance, it would be interesting to find out if users who play games longer are more apt to write positive reviews or if veteran players are more apt to write negative reviews.

Generally, sentiment classification consists of positive, negative and neutral classes (Pang and Lee, 2008). Machine learning techniques like Naïve Bayes, Logistic Regression, Decision Trees and Support Vector Machines (SVM) are heavily used in traditional sentiment analysis models. To classify textual data, these sorts of algorithms have to convert text into numerical representations.

TF-IDF (Term Frequency-Inverse Document Frequency) is one of the most popular feature extraction methods used in sentiment analysis. The TF-IDF value reflects the relative significance of words from a document to a collection of documents. Words that occur with high frequencies but possess high meaning are also given high weighting, and the words that are used a lot are given low meaning. This will help the machine learning models to concentrate on the important keywords, which carry strong sentiment meaning.

Logistic Regression and Support Vector Machine (SVM) are popular machine learning models for sentiment analysis because they are good at dealing with high dimensional textual data. Logistic Regression is a particularly good choice for binary classification problems, as it is efficient and easy to interpret. SVM is recognized as one of the classifiers with high accuracy on text classification tasks since it can effectively partition classes with hyperplanes.

Another popular algorithm used in sentiment analysis is the Naïve Bayes algorithm, which is efficient in terms of computation and has a good performance in using a large amount of textual data. But Naïve Bayes relies on the independence of the features, a assumption which may not hold true and as a result the performance of the Naïve Bayes model may not be as high as more advanced models.

In recent years, the performance of sentiment analysis has been greatly enhanced with the advancement of deep learning techniques. Deep learning models can learn to understand semantic and contextual relationships between text data. The LSTM and BiLSTM networks are used frequently since they can handle the sequential information and the long-term word dependency.

With the rise of natural language processing, transformer-based models like BERT (Bidirectional Encoder Representations from Transformers) and RoBERTa are now the cutting-edge models used for NLP. Devlin et al., 2019, proposed a new model called BERT that leverages bidirectional contextual embeddings to learn the meaning of words from their context in a sentence. BERT-based models can better capture contextual meaning and are more likely to achieve high classification accuracy than traditional machine learning methods, in general.

There are a few games related studies that have used sentiment analysis methods. Sentiment analysis has been employed to assess player satisfaction, to identify toxic behaviour, to forecast game popularity and to tackle issues related to gameplay. One of the most difficult types of gaming reviews is one that involves players using slang and abbreviations, as well as sarcasm and language that is emotionally charged. Despite these obstacles, gaming reviews are extremely helpful in understanding the experiences of gamers and reactions of the community.

One of the most interesting reviews to analyse for FIFA 23 is its mixed ratings from players, which makes it a prime candidate for sentiment analysis. Football immersion, online multiplayer, realistic gameplay, and improved graphics are often cited as the highlights of the positive reviews. Negative reviews, however, often emphasize scripting problems, bugs, server problems, microtransactions and imbalance in gameplay. Sentiment analysis can be used to systematically identify and analyze such patterns.

Text preprocessing is one of the crucial steps in the sentiment analysis process. Preprocessing methods, like converting text to lowercase, removing characters or punctuation, removing stopwords, tokenizing, stemming, and lemmatizing, add value to the quality of text and decrease noise in data sets (Jurafsky & Martin 2021). These processes assist machine learning models to grasp textual info better.

Tokenization breaks up the sentence into smaller parts, typically words. Stopword Removal removes words that are frequently used in English (such as the, is, and and) that tend to not carry any meaningful sentiment information. Lemmatization is the process of converting words to the root form to enable all grammatical forms of the same word to be treated as the same.

Another crucial role that significant to sentiment analysis projects is Exploratory Data Analysis (EDA). EDA enables researchers to gain insights into the nature of their data, locate missing data points, recognize class imbalance, and see how variables interact. Pie charts, histograms, line graphs and word clouds are all popular ways to display insights in a more clear and concise manner.

Word clouds are particularly effective in sentiment analysis for visualising the top words of the most common ones found in reviews. Examples of positive word clouds

include the word "good," "fun," "realistic," and "amazing," among others, while negative word clouds include the word "lag," "broken," "scripted," and "bug," among others.

In sentiment classification, an evaluation measure is crucial in order to assess the performance of the model. Typical measures of evaluation are accuracy, precision, recall, F1-score and confusion matrices. The overall correctness of the model is measured by accuracy; precision and recall gives deeper insights into the quality of classification. F1-score is used when evaluating imbalanced datasets and is a combination of Precision and Recall.

This project builds on the work done before, using the sentiment analysis techniques to analyze FIFA 23 Steam reviews and looking into the relationship between the sentiment of the reviews and the player-related variables like the recent playtime, the number of games owned, the date of the review and the recommendation rating. To obtain meaningful insights from the data, the study employs NLP preprocessing, TF-IDF feature extraction, machine learning classification, and visualization techniques.

4. Methodology

This project aims to use Natural Language Processing (NLP), machine learning, and data visualization to conduct sentiment analysis on FIFA 23 Steam reviews, employing a systematic methodology. The methodology comprises of various steps such as data collection, data pre-processing, exploratory data analysis, feature extraction, sentiment classification, evaluation, and visualization.

This methodology aims to convert Raw textual reviews into valuable insights about players' opinions and their behavior around FIFA 23. The overall workflow of the project will guarantee that the gathered data is cleaned, analyzed, classified and represented in the form of visuals, which will contribute to the research goals and hypotheses.

4.1 Research Design

The method used in this research is Quantitative and Experimental. The project is quantitative because numerical evaluation of textual sentiment data using statistical and machine learning techniques are involved.

The research process consists of several stages:

- Dataset collection
- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature extraction
- Machine learning model development
- Model evaluation
- Interpretation of data and visualization

The study aims to detect sentiment patterns and explore the correlations between the sentiment of the reviews and other player related features like recent playing time, number of games owned, and creation date of reviews.

4.2 Dataset Collection

The dataset used for this project is “FIFA23 Steam Reviews”, which has been downloaded from Kaggle. Kaggle is a popular site that makes data for machine learning and data science research publicly available.

This data set is a collection of user reviews from the Steam gaming platform from the real world. These reviews are from players and their experiences, their opinions and their satisfaction with FIFA 23. The following attributes are important in the data set:

- Review text
- Recommended status (True/False)
- Playtime in the last two weeks
- Total playtime
- Number of games owned
- Number of helpful votes
- Review creation date

The textual and behavioral data are available, which makes the data set appropriate for sentiment analysis and player behavior analysis.

4.3 Data Cleaning and Preprocessing

Websites generally include raw text data that inevitably has noise, inconsistencies, symbols and irrelevant information. As such, the data set should be preprocessed to ensure the quality of the data before performing machine learning analysis. In this project, several pre-processing techniques are used.

4.3.1 Lowercase Conversion

To be consistent across the dataset, all review texts are converted to lower case. That means that words like “Good” and “good” are all treated the same.

4.3.2 Removal of Punctuation and Special Characters

Punctuation marks, symbols, emojis, URLs, and special characters are stripped since they do not generally convey information that is relevant for sentiment classification.

4.3.3 Removal of Numbers

The numbers that are not meaningful for sentiment are discarded from the textual data.

4.3.4 Tokenization

The act of dividing sentences into smaller units, typically individual words, but also sometimes other linguistic elements or phrases, is referred to as tokenization.

Example:

“FIFA 23 gameplay is amazing” → [“FIFA”, “23”, “gameplay”, “is”, “amazing”]

4.3.5 Stop-word Removal

Words that are very common, such as “the,” “is,” “and,” and “a” are excluded as they add little value in sentiment classification.

4.3.6 Lemmatization

The word lemmatization changes words to their base forms.

Example:

The word “playing”, "played" and "plays" are all shortened to the word "play".

This process can decrease redundancy and help in understanding the model. To remove duplicate data or data that is missing, use the Duplicate and Missing Data Removal command.

To enhance data quality and minimize the bias when training models, duplicate rows and rows with missing values are eliminated.

4.3.7 Duplicate and Missing Data Removal

Duplicate reviews and rows containing missing values are removed to improve data quality and reduce bias during model training.

4.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand the characteristics of a data set and to find patterns, before the development of a model. The following analyses are carried out:

- Distribution of positive and negative reviews

- Average player playtime
- Relationship between games owned and sentiment
- Review trends over time
- Frequently used words in reviews

Several visualization techniques are used, including:

- Pie charts
- Bar charts
- Histograms
- Heatmaps
- Word clouds
- Line graphs

Word clouds are used to display frequently occurring positive and negative words found in player reviews.

4.5 Sentiment Labelling – Binary Classification

The `voted_up` column from the Steam reviews is used as the main supervised sentiment label for this project. This label may only have 2 values. True indicates that the player recommended FIFA 23 and False indicates that the player has not recommended the game. Thus, the sentiment classification problem used in this project is considered as a binary classification problem. The following is an interpretation of the labels:

- True = Positive / Recommended
- False = Negative / Not Recommended

Neutral sentiment is not considered as a supervised class since there is not a reliable neutral label in the dataset. While a review can include both neutral or mixed language, Steam will only keep the final recommendation decision: recommended or not recommended. This is why only the `voted_up` label is used to train the machine learning and deep learning models; these models are trained only to predict positive and negative classes.

4.6 Sentiment Labelling – Multiclass Classification

This project also contains a multi-class sentiment classification experiment, in addition to the main binary classification task. This experiment was added to perform written

sentiment analysis in three categories: Positive, Neutral and Negative. The multi-class labels are not directly obtained from Steam, however, as the Steam dataset doesn't contain a neutral ground-truth label.

On the contrary, the multi-class labels are generated based on the compound sentiment scores provided by VADER. The VADER-derived multi-class labels are created using the following thresholds:

- Negative = compound score ≤ -0.05
- Neutral = compound score -0.05 and 0.05
- Positive = compound score ≥ 0.05

So it is not surprising that the binary and multi-class tasks are answering different questions. The binary classification task is based on the actual label “voted_up” to predict the Steam recommendation behaviour. The multi-class classification task, on the other hand, uses weak labels derived from the text of the review to predict the polarity of sentiment in the text, as measured by the VADER scheme. For this reason, the multi-class result is considered as an extra experiment and not the substitute for the primary binary voted_up classification task.

4.7 Feature Extraction

Textual data is not directly useful for machine learning models. Thus, reviews need to be represented by numbers via a feature extraction process first.

In this project, TF-IDF (Term Frequency-Inverse Document Frequency) vectorisation is used.

TF-IDF measures:

- Term Frequency (TF): How often a word appears in a review.
- Inverse Document Frequency (IDF): How unique the word is across all reviews.

TF-IDF helps identify meaningful words while reducing the importance of overly common words.

In other cases, words like “fun,” “amazing,” “broken,” and “lag” might be given more weight owing to their ability to contribute greatly to sentiment meaning. There are various models which employ various representations of text other than TF-IDF.

The BiLSTM neural network is based on learned word embeddings. In this method, the TextVectorization layer is used to process each review into sequences of tokens, and the Embedding layer is trained to extract dense vector representations from these

sequences. These embeddings are not handcrafted, but are learned from the FIFA 23 review set. Contextual embeddings are the method used by DistilBERT for the transformer benchmark. This is because the meaning of a word is dependent on the words that surround it in a sentence.

This is different from TF-IDF. Which is primarily a set of frequency and important words. In this work, DistilBERT is not fine-tuned but is merely used as a pretrained sentiment benchmark.

4.8 Modelling Approaches

In this project, several modelling approaches are used for analyzing FIFA 23 Steam reviews. This project offers a lexicon-based approach to sentiment classification, classic machine learning classifiers, a trained deep learning model, and a pre-trained transformer benchmark, all in one.

These can be done because they are all reviewed from different perspectives of the text. In this project there are the following modelling approaches:

- VADER as the lexicon-based sentiment baseline
- Using traditional machine learning models: Logistic Regression, Linear SVM, Random Forest and Multinomial Naive Bayes.
- The trained deep learning model used is BiLSTM Neural Network.
- DistilBERT SST-2 as the pretrained transformer benchmark

The primary binary classification task is based on the actual Steam voted_up label. The multi-class experiment is additionally performed on the written sentiment data, using weak labels derived from VADER.

4.8.1 Logistic Regression

Logistic Regression is a supervised learning method that is widely used for binary classification problems. In this project, it classifies a review to either positive class or negative class using the TF-IDF (Text features) of a review.

Logistic Regression can be used in sentiment analysis because the text data is often very high dimensional and sparse. It is also efficient and easy to train as well as being relatively interpretable compared to more complex models. For this project, one of the primary supervised sentiment classification baseline models is Logistic Regression.

4.8.2 Linear Support Vector Machine

Linear Support Vector Machine (Linear SVM) is another supervised algorithm which is commonly used for text classification. It does this by searching the feature space for the most optimal separating boundary between positive and negative reviews. The reason why linear SVM is good for sentiment analysis is that the features typically obtained from TF-IDF are high-dimensional.

The linear decision boundary is able to effectively separate sentiment classes. Linear SVM is added to this project as it is used very commonly in classification tasks pertaining to NLP and is frequently having a good performance with review text data.

4.8.3 Random Forest

Random Forest is a technique from the ensemble learning category which utilizes numerous decision trees to create a final classification. Each tree is trained on different patterns of the data, and the final prediction is combined from all trees.

In this project, Random Forest is employed to do a comparison of tree-based learning and linear text classification models. Random Forest can model non-linear patterns, but will not necessarily outperform Logistic Regression or Linear SVM on sparse TF-IDF text features. It is useful as a comparison model, however; as it offers a different learning approach from linear classifiers.

4.8.4 Multinomial Naïve Bayes

Multinomial Naive Bayes is a Naive Bayes classifier that is based on Bayes' Theorem. It works well with word frequency and TF-IDF features and is often used for text classification.

This model is fast and efficient for dealing with large collections of text since it assumes that features are conditionally independent of one another. But this might restrict its performance as words in reviews can be related to each other. In this project, Multinomial Naive Bayes is considered as a simple and efficient baseline model for sentiment classification.

4.8.5 Transformer DistilBERT SST-2

A pretrained DistilBERT SST-2 transformer model is also used as a deep learning benchmark. The FIFA 23 Steam review dataset is not used to train or fine-tune the

DistilBERT model in this project, unlike the BiLSTM model. Rather, it is used in Hugging Face Transformers pipeline to show the difference between a general pretrained sentiment prediction and the Steam voted_up recommendation label.

DistilBERT is a contextual word representation-based model, able to understand words in the context of their surrounding words. The model was trained on general sentiment data and may not be able to pick up on Steam gaming slang, sarcasm, mixed sentiments, and/or recommendations. Hence, the pretrained transformer model, DistilBERT, serves as a benchmark while the trained deep learning model is the BiLSTM.

4.8.6 BiLSTM Neural Network

This project introduces an additional trained deep learning model, a Bidirectional Long Short-Term Memory (BLSTM) neural network. The BiLSTM model is a model without the use of TF-IDF features, as opposed to traditional machine learning models. Instead, it transforms the text of the review into an embedded layer composed of dense numerical vectors and a text vectorization layer, which transforms the review text into sequences of tokens.

The BiLSTM network consists of TextVectorization layer, Embedding layer, Bidirectional LSTM layer, Dropout layers, Dense hidden layer and a Sigmoid output layer. With sigmoid output, the model predicts the probability of the review being positive or recommended, so it is appropriate for binary classification. The model is trained on the basis of binary cross-entropy loss and the Adam optimiser.

To mitigate overfitting, a validation split and early stopping were employed in the training process. The validation split enables the model to track performance on the validation data set as training progresses, and early stopping prevents overfitting by stopping training when the validation error starts to rise. This is what makes the BiLSTM a real trained deep learning model as it learns directly from the cleaned FIFA23 Steam review dataset.

4.9 Training and Testing Process

80:20 split of cleaned dataset into training and testing sets. The training set is used to train the supervised models and the testing set is used to test the models on unseen reviews.

A split stratified by class distribution is used to ensure the training set and test set remain similar in distribution of classes. This makes it significant because the binary dataset has 9,310 positive reviews and 10,740 negative reviews. The class difference wasn't too big so SMOTE was not used. SMOTE is more appropriate when there is a significant

imbalance in one particular class. The SMOTE method might also generate more complex patterns of the text features, which are difficult to interpret when applied to the sparse TF-IDF features used in this project.

However, stratified splitting, F1 score, and confusion matrix is used to fairly assess model performance. In this project, Grid Search was not used. Fixed and reasonable hyperparameters were used in the training of the models to make the experiment feasible within the scope of the assignment and runtime within Colab. The TF-IDF, for instance, includes unigrams and bigrams with a maximum of 10,000 features, while the BiLSTM has a fixed vocabulary size, sequence length, embedding dimension, batch size and early stopping.

Future work is recommended to carry out more detailed hyperparameter tuning. The cleaned review text is first vectorized using TF-IDF to numerical features for the traditional machine learning models. These TF-IDF features are then passed into Logistic Regression, Linear SVM, Random Forest and Multinomial Naive Bayes models to be trained. The review text goes through a TextVectorization layer in the BiLSTM neural network.

The text is tokenized into a sequence of tokens, fed into an Embedding layer, and learned using a Bidirectional LSTM layer. To prevent over fitting dropout layers are implemented, and early stopping is implemented with a validation performance. The transformer model is different. We use a pretrained DistilBERT SST-2 sentiment model for the text of the review. These predictions are then compared to Steam voted_up label to assess the accuracy of a general pretrained model to match FIFA 23 recommendation labels.

For the multi-class experiment, the same cleaned review text is used, but the target label is changed from binary voted_up to the VADER-derived Positive, Neutral, and Negative label. In the traditional machine learning models, TF-IDF features are used while in the multi-class BiLSTM model text vectorization, embeddings, and Bidirectional LSTM layer are used. The BiLSTM output layer changes from sigmoid output to softmax output with three units for Negative, Neutral and Positive classes. The loss function is also modified from binary cross-entropy to multi-class or sparse multi-class cross-entropy for multi-class classification.

The evaluation of the multi-class models is performed on the basis of accuracy, macro F1-score and weighted F1-score. Macro F1-score is crucial since it considers all three sentiment categories equally. Additionally, the weighted F1 score is reported since it takes into account the number of reviews in each class. This makes a fair comparison of models when measuring the categories of written sentiment: Positive, Neutral and Negative.

4.10 Model Evaluation

To assess the effectiveness of the sentiment classification models, they are evaluated using different performance metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Accuracy is the overall percentage of correct predictions. Precision is the number of positive reviews that are correctly predicted.

Recall: the number of correct positive reviews identified. When the data set includes positive and negative reviews, F1 score integrates precision and recall into a single "balanced score" and can be used to compare models.

The number of correct and incorrect predictions for each class is also displayed using confusion matrices. This can help you determine if a model is more likely to predict positive or negative reviews or both.

Traditional machine learning models are also applied with the 5 fold stratified cross validation as a stability check for selected models. Cross validation is used to help demonstrate that a model will perform well on a different split of the data, rather than relying on just one train-test split.

We report the mean F1 score and the standard deviation in the cross validation results. Evaluation of models is done with a consideration for overfitting. In traditional machine learning models, cross-validating to see if performance is constant. To prevent

overfitting, validation split, dropout layers and early stopping are implemented in the BiLSTM model. The training history plots are also analyzed to make comparisons between training and validation behaviour.

For the final model comparison, Logistic Regression, Linear SVM, Random Forest, Multinomial Naive Bayes, BiLSTM Neural Network and Transformer DistilBERT SST-2 models were included. This enables the project to measure the performance of traditional supervised machine learning models, a trained deep learning neural network, and a pretrained transformer on the same metrics.

4.11 Data Visualization and Interpretation

Findings are clearly presented using visualization techniques, which enhance the understanding of patterns of sentiment. The visualizations include:

- Sentiment distribution charts
- Correlation heatmaps
- Review trend analysis over time
- Playtime versus sentiment graphs
- Word clouds for positive and negative reviews

These visualizations provide valuable insights into the most prevalent opinions, game problems, and general sentiments of the player community regarding FIFA 23.

4.12 Tools and Technologies

The project is executed in the Google Colab environment, with Python code. The following libraries and technologies are used:

- Pandas – Data manipulation and preprocessing
- NumPy – Numerical operations
- NLTK – Text preprocessing
- Scikit-learn – Machine learning models
- Matplotlib – Data visualization
- Seaborn – Statistical visualization

CDS6344 – Social Media Computing

- WordCloud – Word cloud generation
- Google Colab – Cloud-based development environment
- VADER – Lexicon-based written sentiment scoring
- SciPy – Statistical hypothesis testing
- Hugging Face Transformers – Pretrained DistilBERT SST-2 sentiment model
- TensorFlow/Keras – Deep learning model development and BiLSTM training

GitHub and Google Colab are also used for collaboration, code storage and project documentation.

5. Sentiment Analysis

The sentiment analysis phase is the stage that concentrates on comprehending the general trend of recommendations in FIFA 23 Steam reviews. In this project, the main supervised sentiment label is the Steam voted_up column. A value of True is interpreted as a positive recommendation and a value of False is interpreted as a negative recommendation.

This label is great as it directly promotes whether the reviewer would recommend it to other Steam users. It must however be understood as a cautionary message signal, not as an ideal indicator of emotional sentiment. The main classification task is binary, as there is a binary supervised label available. Labelling is not provided by steam in this dataset. Some review texts might be neutral or mixed, but the final Steam recommendation will be categorized as recommended or not recommended.

Thus, the neutral sentiment is not considered as a supervised class for the machine learning and deep learning models. Once the data was cleaned 20,050 reviews were left for analysis. It has 9,310 positive reviews and 10,740 negative reviews. This indicates that the review distribution is slightly skewed towards negative reviews. The disparity indicates that during the period examined, the football world has experienced a split response to FIFA 23 on Steam, with a prevailing trend of discouraging the game.

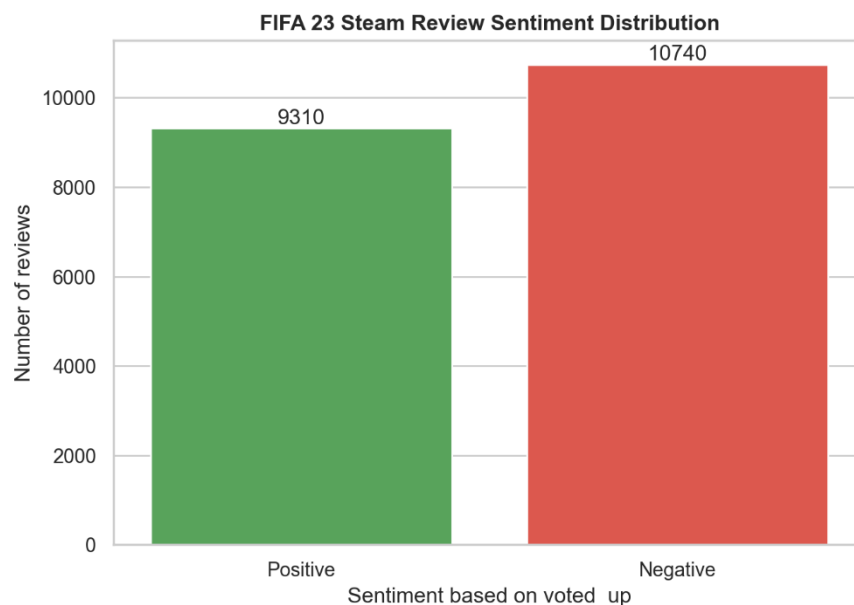


Figure 5.1: Steam Review Sentiment Distribution

VADER sentiment analysis is also used as a lexicon-based comparison approach to the text of the review. Normally, VADER will return a compound sentiment score that can be interpreted with three classes: positive, neutral, and negative. For this project, the following rule was used: positive if compound score > 0.05 , neutral if compound score $-0.05 < \text{compound score} < 0.05$, negative if compound score < -0.05 . With this 3-class VADER interpretation, 35.75% of the cleaned reviews are considered to be positive reviews.

There are also 31.04% more (6,224) reviews that are classified as neutral and 33.21% more (6,659) that are classified as negative. This descriptive result demonstrates that FIFA 23 reviews on Steam are better than the written tone of FIFA 23 reviews. While Steam only records the final recommendation as positive or negative, there are many reviews which include neutral or mixed language.

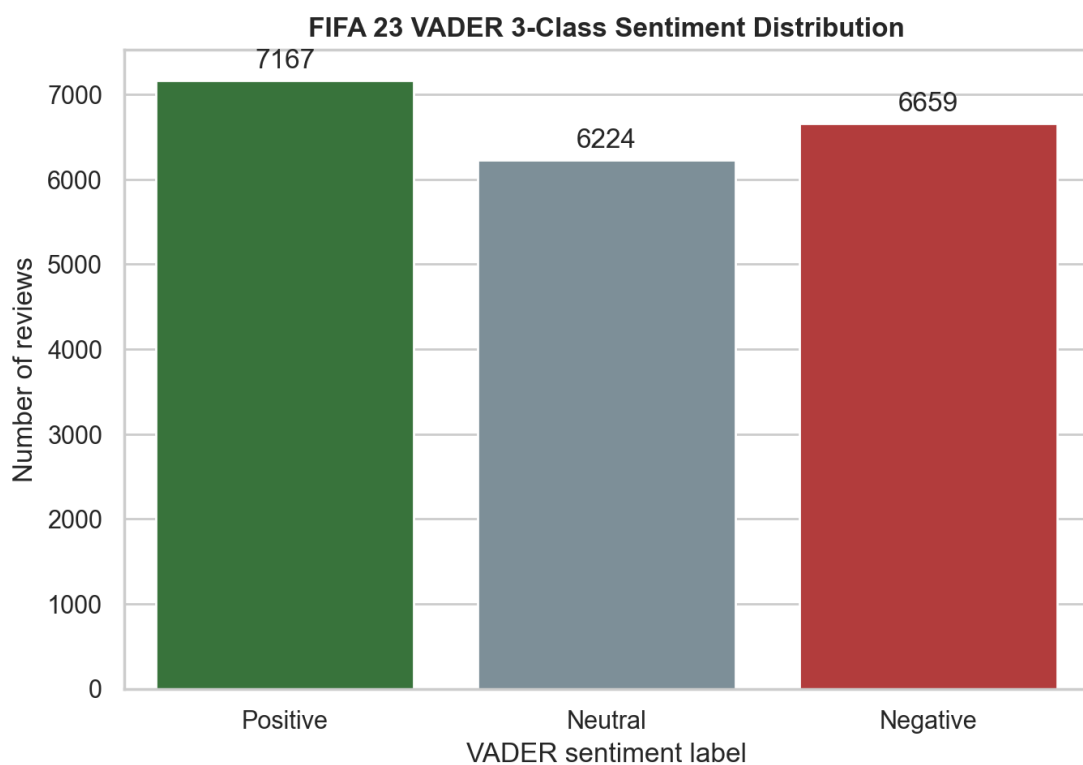


Figure 5.2: VADER 3-class sentiment distribution before binary conversion

VADER needs to be re-trained as a binary label for Hypothesis 4 since it is being compared directly to Steam voted_up label. A confusion matrix needs the labels compared to have the same class structure. Voted_up has no neutral class, so VADER will be converted following binary rule: positive if the compound score is greater than or equal to 0; and negative if the compound score is less than 0.

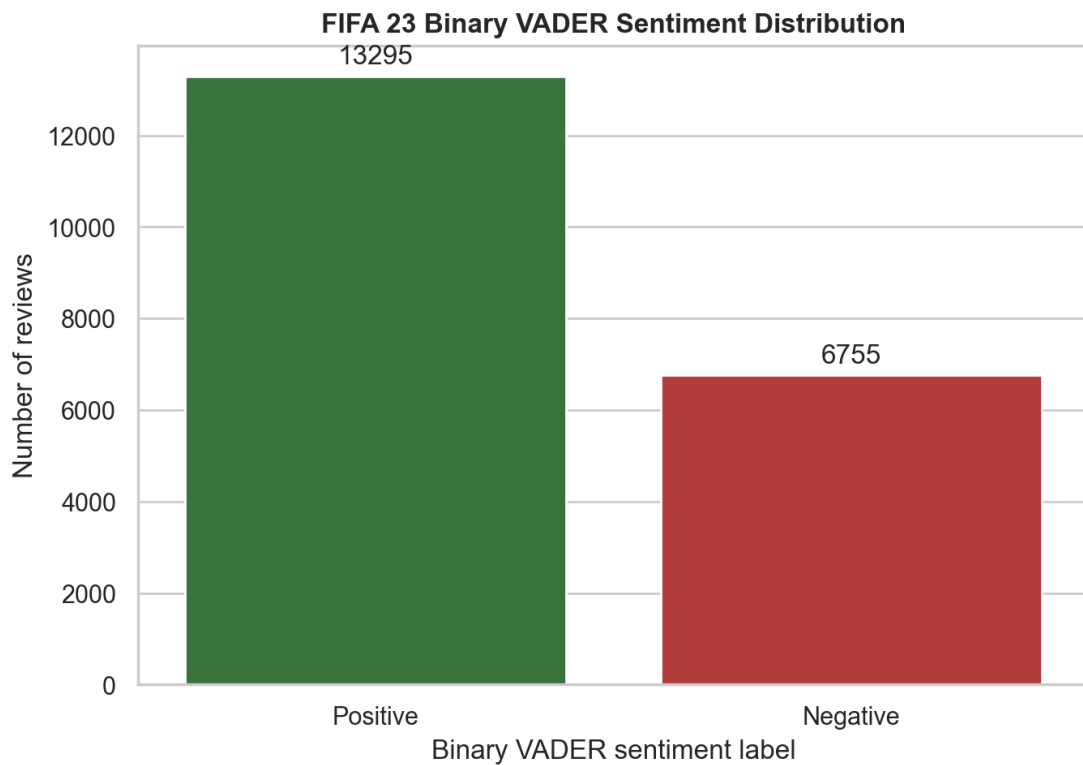


Figure 5.3: Distribution of binary VADER sentiment labels after conversion for comparison with Steam

The Lexicon-based written sentiment labels are displayed in a binary VADER distribution without the Neutral category, so they can be directly compared to the binary Steam voted_up recommendation label. The agreement rate for the comparison is 69.36%. That is, the sentiment of the review is generally in line with the Steam recommendation when it is written. But, in about one-third, there is no match. These are mismatches and will occur as the behaviour of Steam to recommend videos is not always the same as what people actually write.

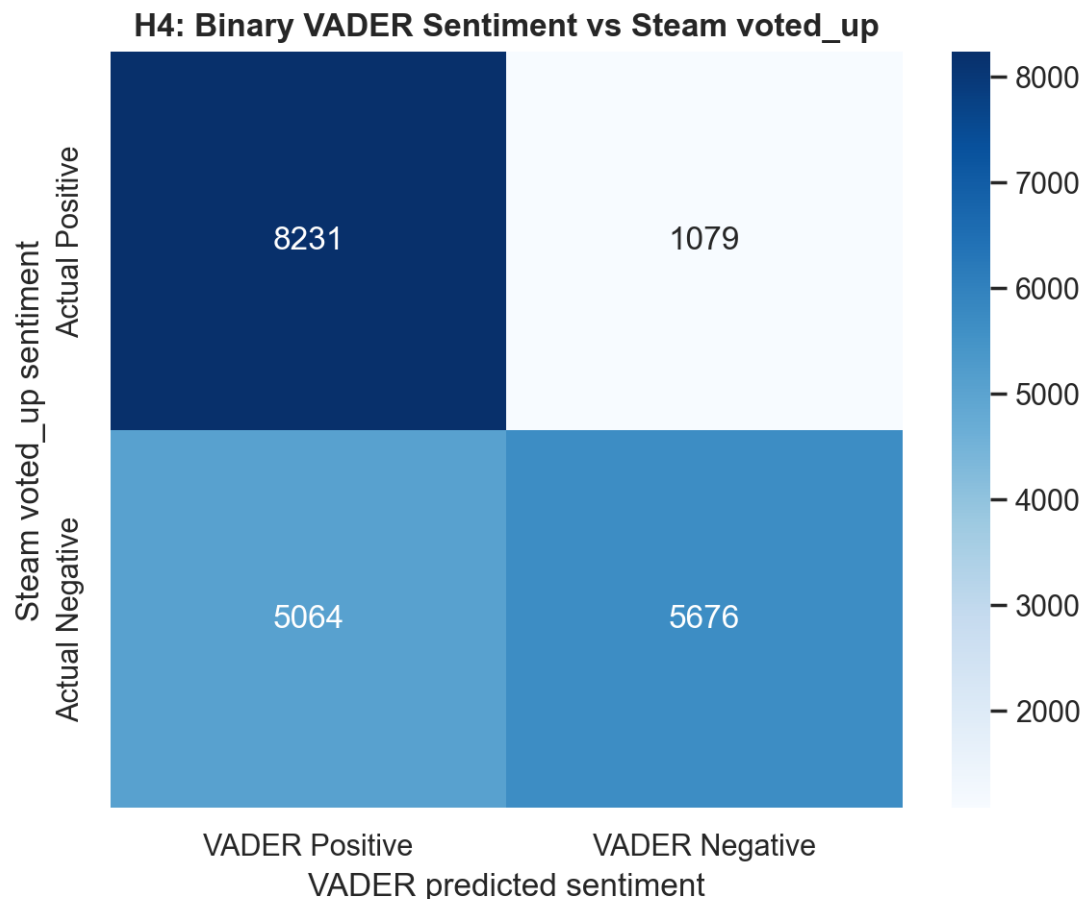


Figure 5.4: Confusion matrix comparing binary VADER sentiment with Steam voted_up recommendation labels.

Some may be positive but still not recommend the game due to bugs, anti-cheat issues or online problems, or price. Other players might write short, sarcastic, neutral or mixed reviews that a lexicon-based method is unable to perfectly interpret. So, the VADER comparison (Hypothesis 4) is only partly supported: There is some degree of correspondence between written sentiment and Steam recommendation, but they are not exactly the same.

The labels of the target words are also generated using the VADER 3-class distribution for the additional multi-class sentiment classification experiment. There are 6,659 Negative reviews, 6,224 Neutral reviews and 7,167 Positive reviews in the distribution. These percentages are 33.21%, 31.04% and 35.75% of the cleaned data set. This indicates that the written sentiment labels derived from the VADER are fairly balanced among the three classes.

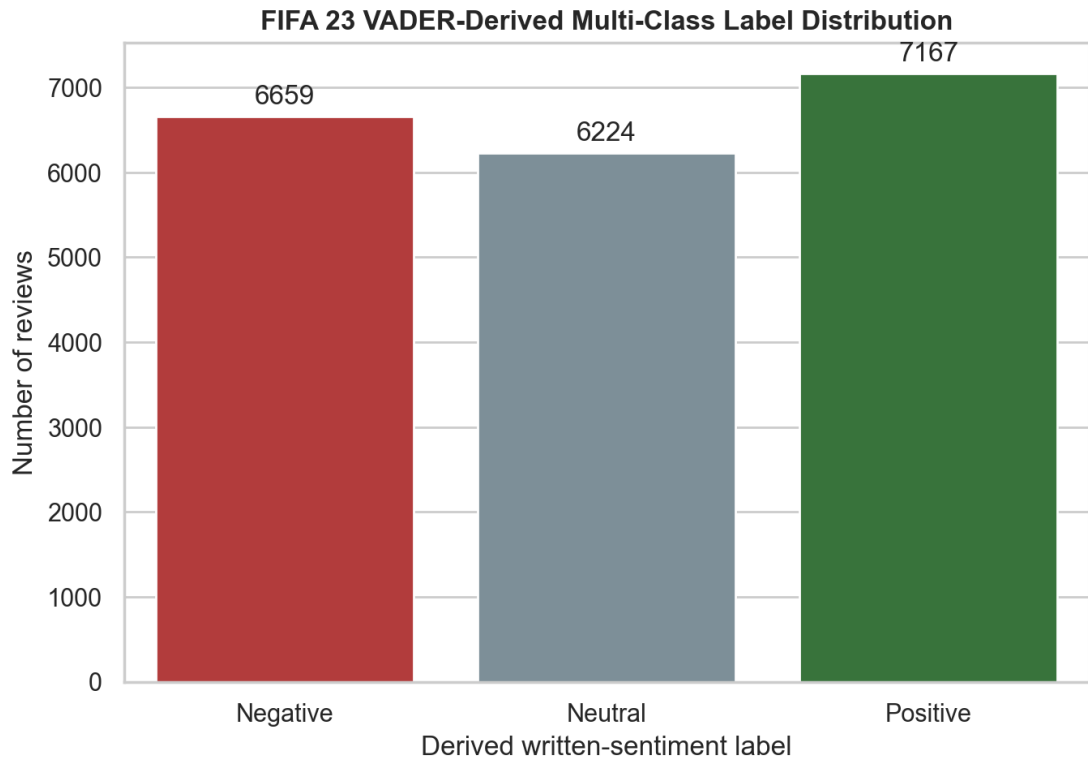


Figure 5.5: Distribution of VADER-derived multiclass sentiment labels

These multi-class labels should, however, be used with caution. Not human-annotated labels but weak labels obtained from a lexicon-based sentiment technique. Thus, they are not exactly VADER polarity, but rather human-labelled emotions. Unlike the binary `voted_up` task, where the model's prediction is the actual result of a Steam recommendation, this task is predicting the recommendation outcome itself. The binary task is the primary supervised task, as it takes into account the actual label that has been provided in the data; the multi-class task is added for comparison so that it can be seen how the different models perform when adding the Neutral sentiment.

6. Transformers / Deep Learning Models

This section provides an explanation of the deep learning and transformer based aspects of the project. A trained BiLSTM Neural Network was incorporated after feedback from the tutor to meet the deep learning requirement. The DistilBERT SST-2 model is not considered as the primary fine-tuned deep learning model since it was not trained on the FIFA 23 Steam review data set.

The actual deep learning model used in this project is a BiLSTM Neural Network. It is fine-tuned directly on the cleaned FIFA 23 Steam reviews with the `voted_up` binary label. The model includes a TextVectorization layer to transform the text from reviews into numerical sequences of tokens, which is then followed by an Embedding layer to learn dense representations of words. A Bidirectional LSTM layer is then used to learn sequential text patterns from both forward and backward directions. This can be helpful when revising reading text since the context of a word can be determined by the context of adjacent words that precede and/or follow it.

In addition to the BiLSTM architecture, the model comprises of Dropout layers and Dense hidden layer. Randomly disable a portion of the neurons during training to reduce overfitting, known as dropout. The final layer has a sigmoid activation function since the problem is a binary classification problem. The model is trained on binary cross entropy loss with Adam optimizer. Early stopping is implemented by splitting the data into a validation set and stopping when the performance no longer improves on the validation set.

The accuracy, precision, recall, and F1-score of the BiLSTM Neural Network were 0.8404, 0.8479, 0.7997, and 0.8231, respectively. This is because the BiLSTM has acquired helpful sentiment patterns from the text of the review. It is not as good as Logistic Regression, but better than other models: Random Forest, Multinomial Naive Bayes, and pretrained DistilBERT SST-2 benchmark. This indicates that the model is a valid trained deep learning model for the project.

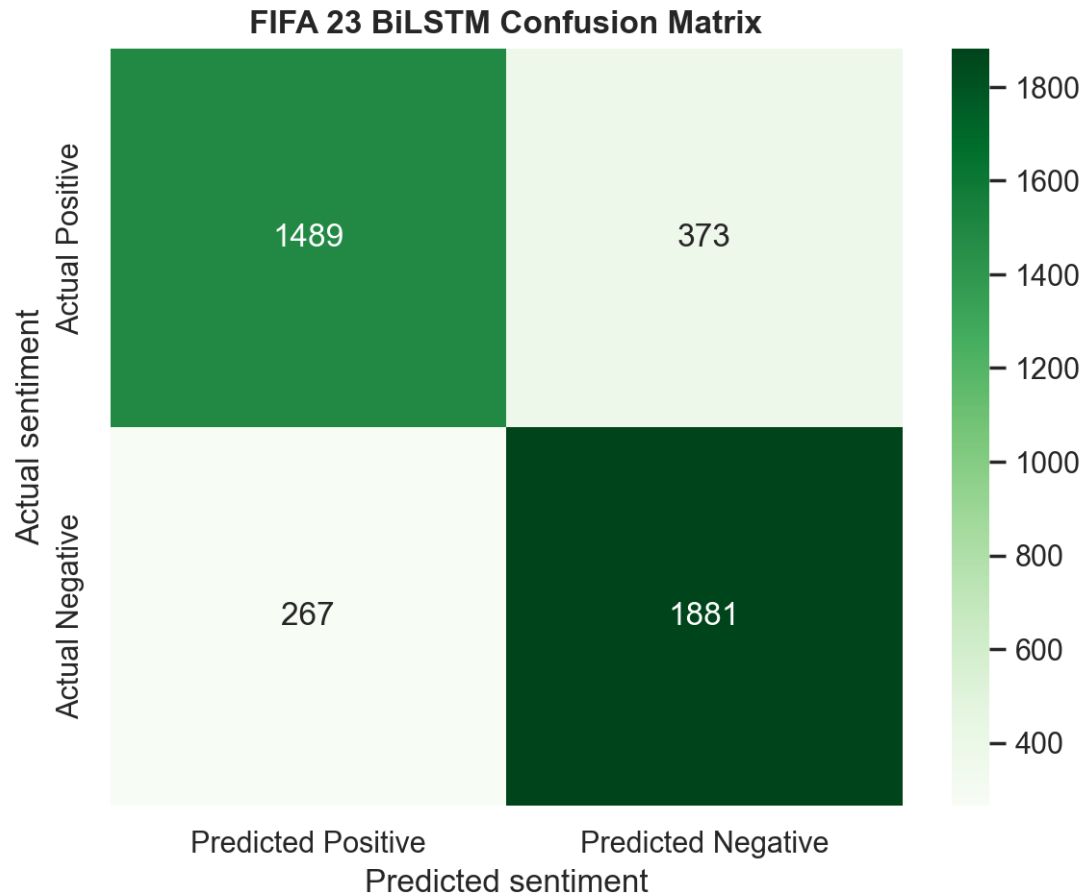


Figure 6.1: Confusion matrix for the trained BiLSTM neural network on the FIFA 23 Steam review test set.



Figure 6.2: Training and validation loss/accuracy history for the BiLSTM neural network.

The BiLSTM model is also fine-tuned for the multi-class sentiment classification experiment. The BiLSTM has a sigmoid output layer in the binary task that determines the sentiment of a review, defined by the Steam voted_up label. In the multi-class task, the output layer is modified to be a softmax layer having three output units: Negative,

Neutral, and Positive. This enables the model to make a prediction for one of the three VADER-derived sentiment categories.

The multi-class BiLSTM model gave the best performance with an accuracy of 0.8429, macro F1-score of 0.8450 and weighted F1-score of 0.8436. It can be seen that the BiLSTM is able to learn useful patterns from the review text for written sentiment classification with three classes. The labels used in this experiment however are generated from the compound scores of the VADER models and the results should be interpreted as the model's performance in reproducing VADER polarity labels and not as real human labelled sentiment.

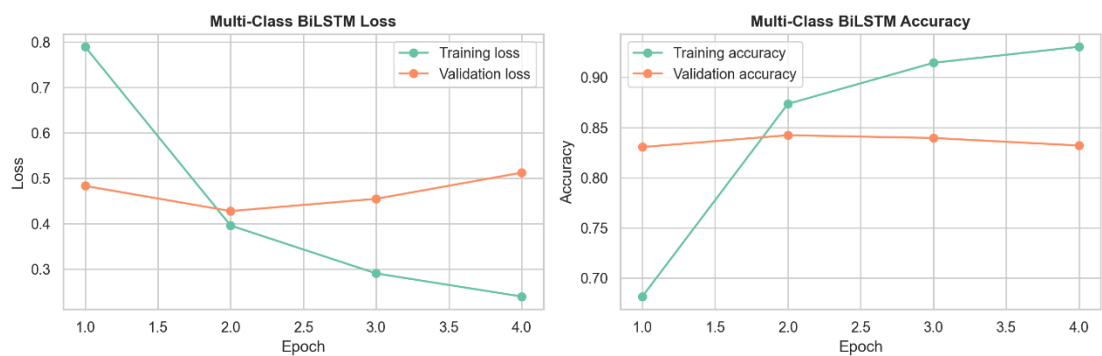


Figure 6.3: Training and validation history for the multi-class BiLSTM model.

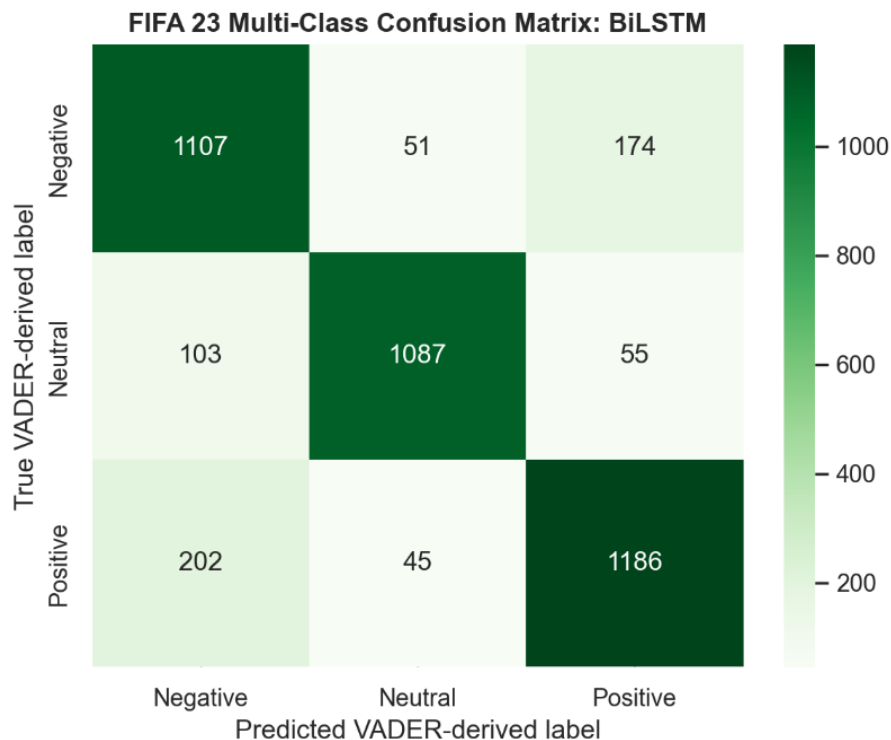


Figure 6.4: Confusion matrix for the multi-class BiLSTM neural network.

DistilBERT SST-2 model has been pretrained as a transformer benchmark. Unlike the BiLSTM model, it is not the FIFA 23 Steam data set that it is trained on. Rather, it is used via the Hugging Face Transformers pipeline. DistilBERT is trained with general sentiment data and can capture general sentiment patterns, but it might not accurately reflect Steam review behavior since it was not trained on Steam-specific labels for reviews.

The DistilBERT SST-2 model achieved an accuracy of 0.7565, precision of 0.8184, recall of 0.6114, and F1-score of 0.6999. The fact that it has lower performance doesn't mean transformer models are weak. Rather, it demonstrates the need for domain adaptation. Some slang, sarcasm, mixed feelings, tech complaints, and recommendations are not always in emotional terms are common in steam reviews. Also, voted_up is not a comment on all sentences in the review, but whether the player recommends FIFA 23.

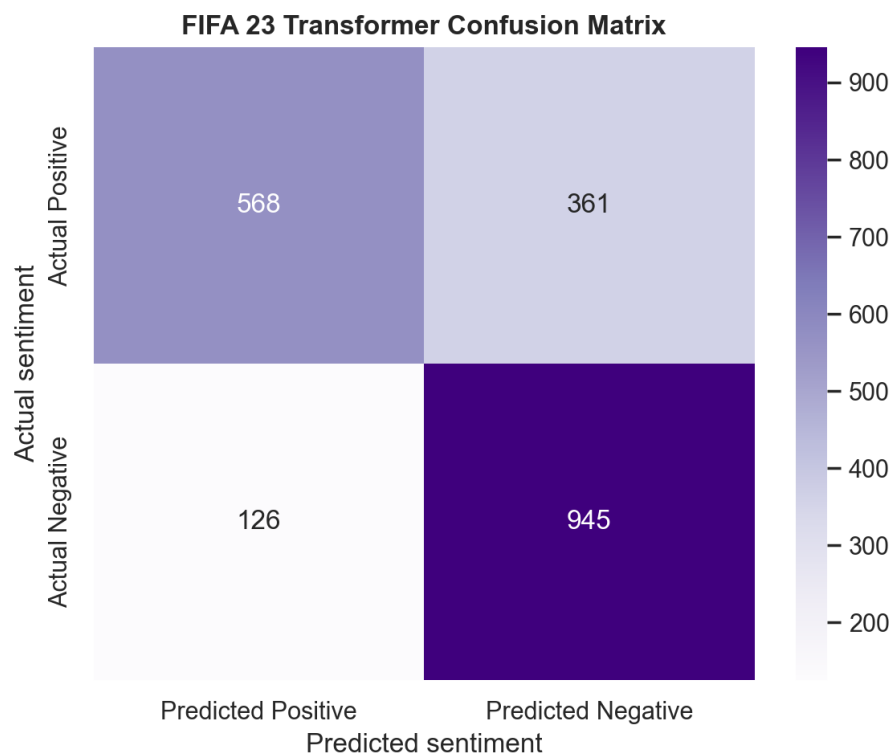


Figure 6.5: Confusion matrix for the pretrained DistilBERT SST-2 benchmark against Steam voted_up labels.

In conclusion, the BiLSTM model supplies the trainable deep learning part for the project, and DistilBERT supplies a valuable benchmark for pretrained transformer models. As seen from the comparison, deep learning can learn meaningful patterns of reviews, but by a lot of training and designing labels of the specific domain.

7. Result & Visualization

In this section, the key results obtained from the sentiment analysis, hypothesis testing, machine learning models, opinion mining; and aspect-based sentiment analysis are shared. The results are shown to highlight that there are slightly more negative than positive recommendations in FIFA 23 Steam reviews, with 10,740 negative reviews in the cleaned dataset and 9,310 positive reviews in the cleaned dataset. This indicates that the overall player response on Steam was mixed but, leaned more toward dissatisfaction.

The traditional machine learning results indicate that Logistic Regression has the highest overall binary classification accuracy of 0.8566 and the F1-Score of 0.8468. Linear SVM was the second-best model, with an F1-score of 0.8318. It is noted that the trained BiLSTM Neural Network obtained an F1-score of 0.8231, whereas the other classifiers obtained lesser F1-scores. The F1-score of the pre-trained DistilBERT SST-2 transformer benchmark model was lower at 0.6999 when compared to the supervised TF-IDF models and trained BiLSTM model.

Table 7.1: Performance comparison of traditional ML, trained BiLSTM deep learning, and pretrained DistilBERT transformer models.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.8566	0.8403	0.8534	0.8468
Linear SVM	0.8414	0.8193	0.8448	0.8318
BiLSTM Neural Network	0.8404	0.8479	0.7997	0.8231
Random Forest	0.8155	0.7802	0.8389	0.8085
Multinomial Naive Bayes	0.7793	0.8621	0.6246	0.7244
Transformer DistilBERT SST-2	0.7565	0.8184	0.6114	0.6999

Table 7.2: Five-fold stratified cross-validation results for selected traditional machine learning models.

Model	Mean CV F1-score	Standard Deviation
Logistic Regression	0.8402	0.0062
Linear SVM	0.8356	0.0046
Multinomial Naive Bayes	0.7087	0.0105

The cross validation results agree with the main train-test results. Linear SVM is the second best model in terms of mean cross validation F1 score after Logistic Regression.

The values in standard deviation are low, indicating the model performance is quite stable in the five folds. This decreases the chance that the end product will occur through only one chance train-test split.

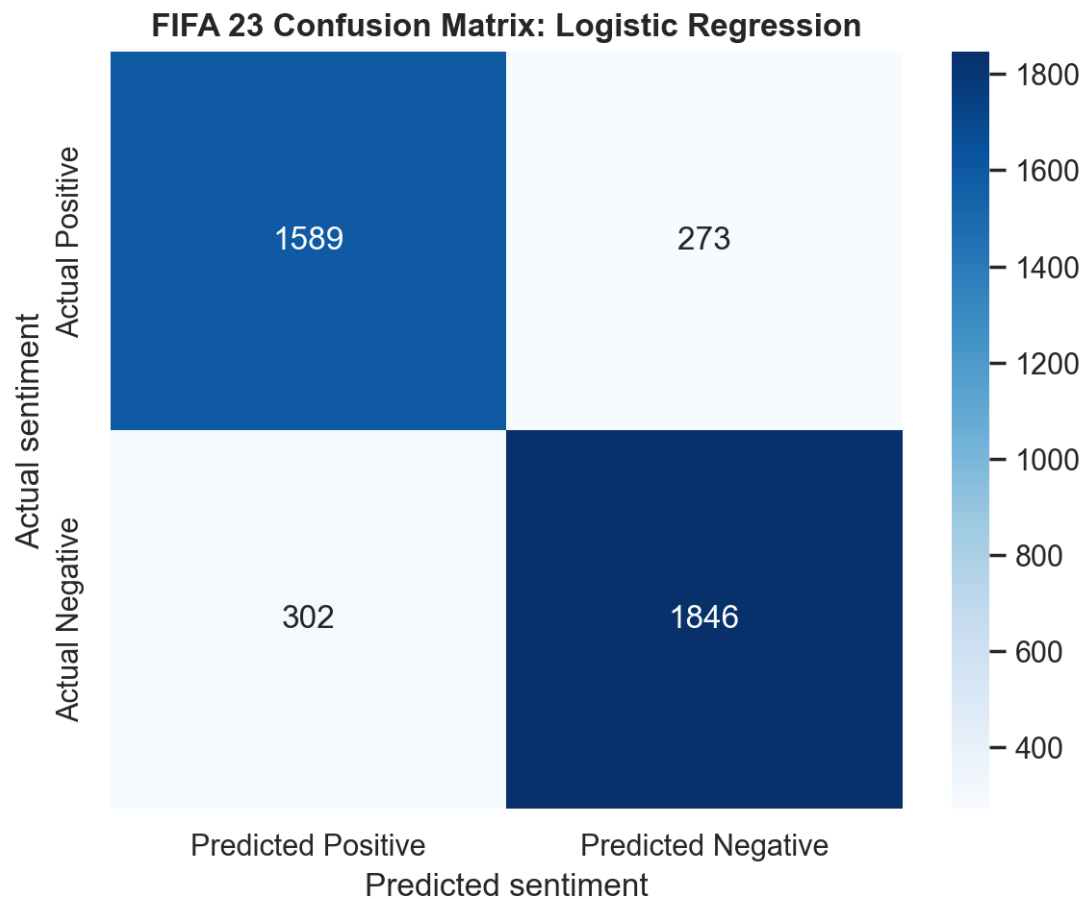


Figure 7.1: Confusion matrix for the best binary supervised model, Logistic Regression.

The results of this trial indicate that TF-IDF features in Logistic Regression still are the best choice in this data set. BiLSTM Neural Network also yields excellent results, indicating that a trained deep learning model is able to learn useful patterns from FIFA 23 review text. The F1 score is lower than that of Logistic Regression, however, this indicates that the data and task is suitable for simpler linear text classifiers. The results show that the benchmark model, DistilBERT, is the worst performing model compared, as it was not fine-tuned on FIFA 23 Steam reviews or the voted_up label (recommendation).

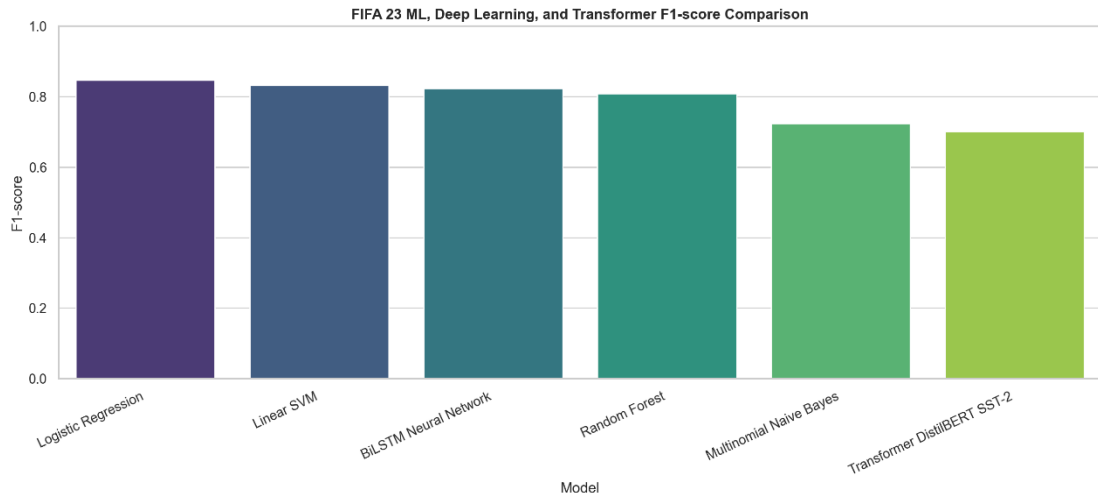


Figure 7.2: F1-score comparison across traditional ML, trained BiLSTM deep learning, and pretrained DistilBERT transformer models.

The results of hypothesis testing are also useful behavioural insights. H1 is accepted, as players who had positive reviews would have a higher median Play time in last 30 days (5.07 hours), than those with negative reviews (0.32 hours). This indicates that those who were "new" to the game were more likely to recommend it.

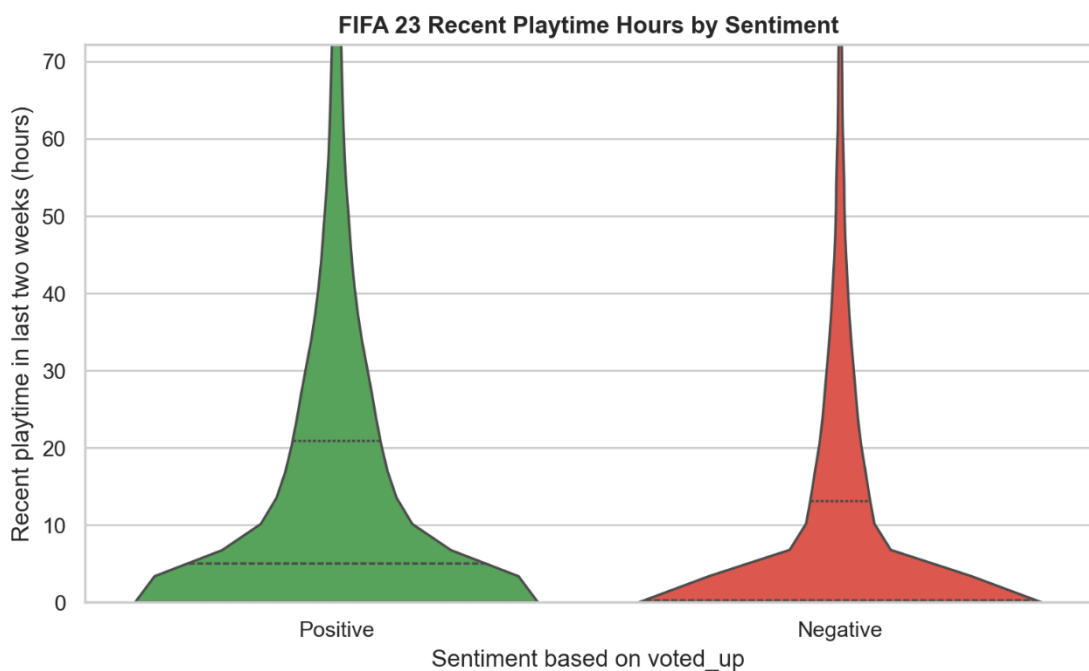


Figure 7.3: Recent playtime hours by Steam review sentiment.

Hypothesis 2 is also supported since there were more games owned by negative reviewers (30 games) than positive reviewers (16 games). This could be a sign that more experienced gamers are more critical of their evaluation.

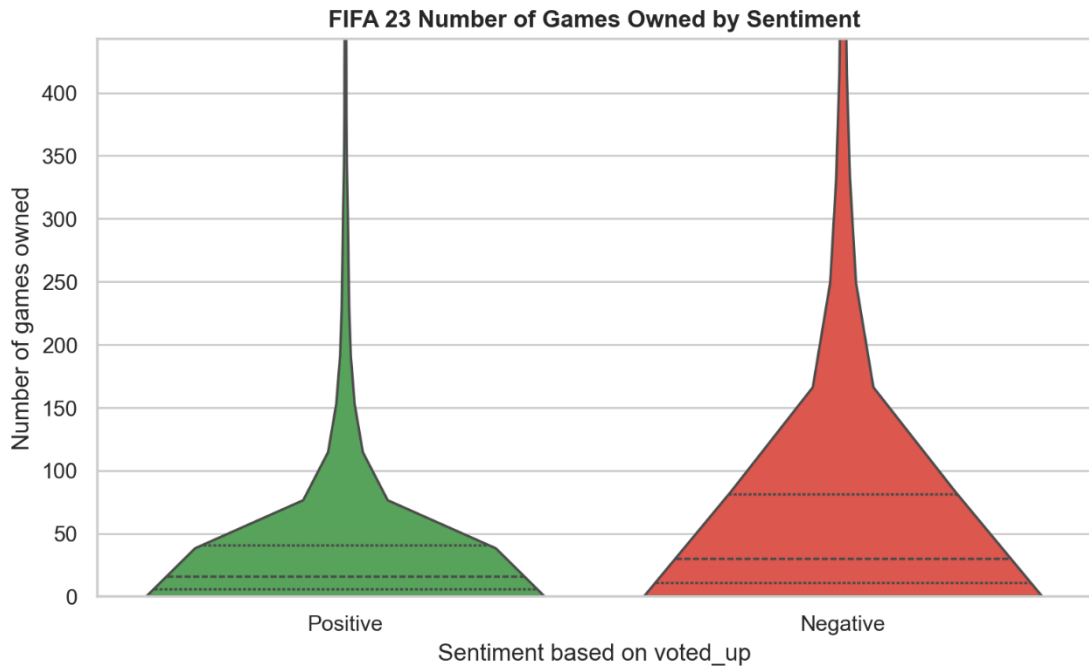


Figure 7.4: Number of games owned by Steam review sentiment.

The final hypothesis is supported (H3) as the chi-square test indicates that sentiment has had a significant change over time. The monthly trend indicates that the percentage of positive reviews at any given month in the review period ranged from September 2022 through March 2023. The drop in player sentiment may be due to several factors since the game's release, such as updates, bugs, service quality, or evolving expectations.

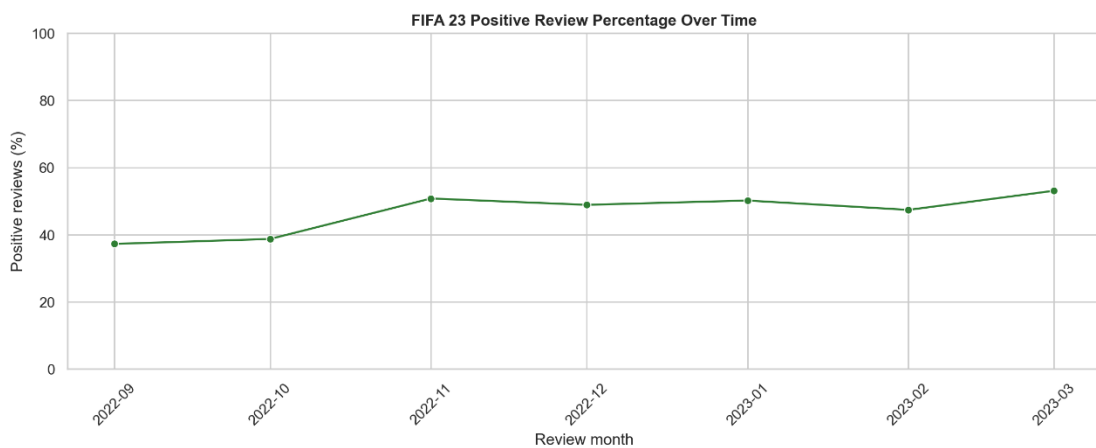


Figure 7.5: Average sentiment trend of FIFA 23 Steam reviews by month.

Hypothesis 4 partially supported. For the cleaned reviews, VADER written sentiment was most similar to the Steam voted_up label, being correct 69.36% of the time. This result visualises that there is a correlation between the written sentiment and the recommendation behaviour, but not a full exact match. The conflict between the two

labels will be predicted as Steam reviews tend to be sarcastic, have mixed feelings, have brief remarks, and may even have complaints about specific issues, despite the language used in the general part of the review being positive.

The VADER vs voted_up confusion matrix offers a better understanding of the areas of agreement or disagreement between written sentiment and Steam recommendation behaviour. Correct matches will be based on the review being classified as positive by VADER and classified as positive by voted_up or classified as negative by VADER and classified as negative by voted_up. Mismatches are when the written tone is positive, but the user gives a negative recommendation, or when the written tone is negative, but the user gives a positive recommendation. These mismatch cases are interesting as they indicate that the decision of Steam recommendation is not necessarily aligned with the emotional tone of sentences.

VADER was also tested prior to binary conversion with its default 3-class sentiment interpretation. There are 7167 positive reviews, 6224 neutral reviews, and 6659 negative reviews in the 3-class VADER distribution. This indicates that there is a significant neutral part in the text of the written review, as recognized by VADER. But it is not suitable to be used directly as a supervised class because there are only positive or negative recommendation outcomes in the Steam voted_up label. Hence, the 3-class VADER is employed descriptively, and the binary VADER is employed directly to compare with voted_up.

Table 7.3: Statistical hypothesis testing results for player behavior and sentiment patterns.

Hypothesis	Test	Main Result	Interpretation
H1: Higher recent playtime relates to positive reviews	Mann-Whitney U	$p = 5.921e-121$; positive median = 5.07 hours; negative median = 0.32 hours	Supported
H2: More games owned relates to negative reviews	Mann-Whitney U	$p = 7.634e-187$; positive median = 16; negative median = 30	Supported
H3: Sentiment changes over time	Chi-square	$p=7.256e-43$; sentiment distribution changes significantly over time.	Supported

H4: Written sentiment agrees with <code>voted_up</code>	VADER agreement	VADER binary sentiment agrees with <code>voted_up</code> for 69.36% of cleaned reviews. Mismatches may indicate sarcasm, slang, short text, or mixed opinions.	Partially supported
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The positive opinions include terms like “good game” and “best game,” “great game”, “ultimate team”, “nice game”, “next gen”, and “amazing game”. The above-mentioned terms indicate that the positive reviewers tend to emphasise the enjoyment, the football experience, and the game modes. Negative reviews contain the following phrases: launch game, anti cheat, worst game, shit game, open game, able play, bad game, fix game, cant play. These sentences reveal that a lot of the complaints are related to launch issues, technical problems, anti-cheat problems and even gameplay problems.

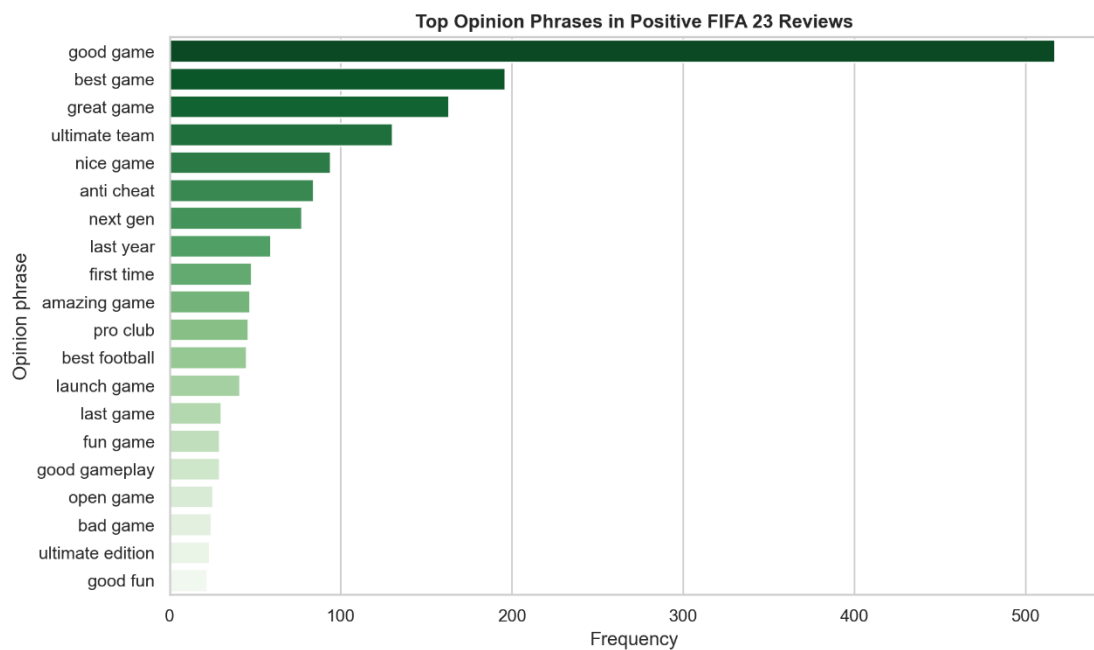


Figure 7.6: Top positive opinion phrases in FIFA 23 Steam reviews.

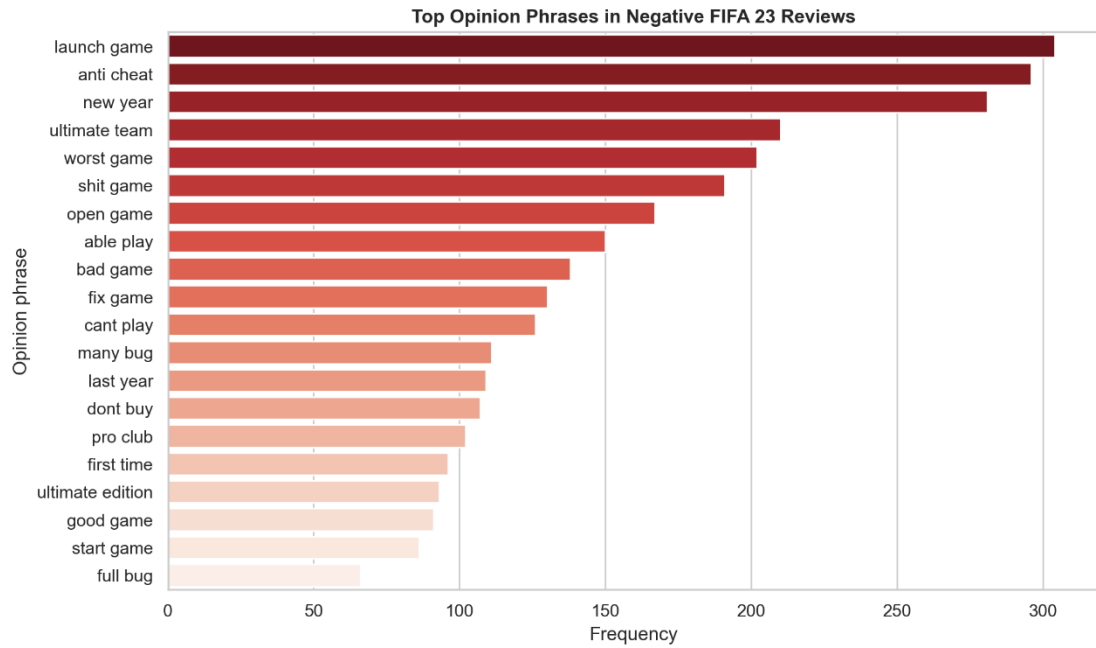


Figure 7.7: Top negative opinion phrases in FIFA 23 Steam reviews.



Figure 7.8: Word cloud of frequent terms in positive FIFA 23 Steam reviews

The aspect-based sentiment analysis is the most straightforward for understanding players' likes and dislikes. Price/Monetization received the highest number of aspect mentions, with 4,207 mentions, and it was strongly negative, with 84.22% negative sentiment. The negative sentiment was also quite high for Servers/Online at 83.22% negative sentiment vs 80.05% for Performance/Bugs.

Table 7.4: Aspect-based sentiment results for FIFA 23 Steam reviews.

Aspect	Total Mentions	Positive %	Negative %
Gameplay	2,197	33.96%	66.04%
Graphics	752	56.25%	43.75%
Performance/Bugs	2,241	19.95%	80.05%
Servers/Online	721	16.78%	83.22%
Career/Content	1,769	35.56%	64.44%
Price/Monetization	4,207	15.78%	84.22%

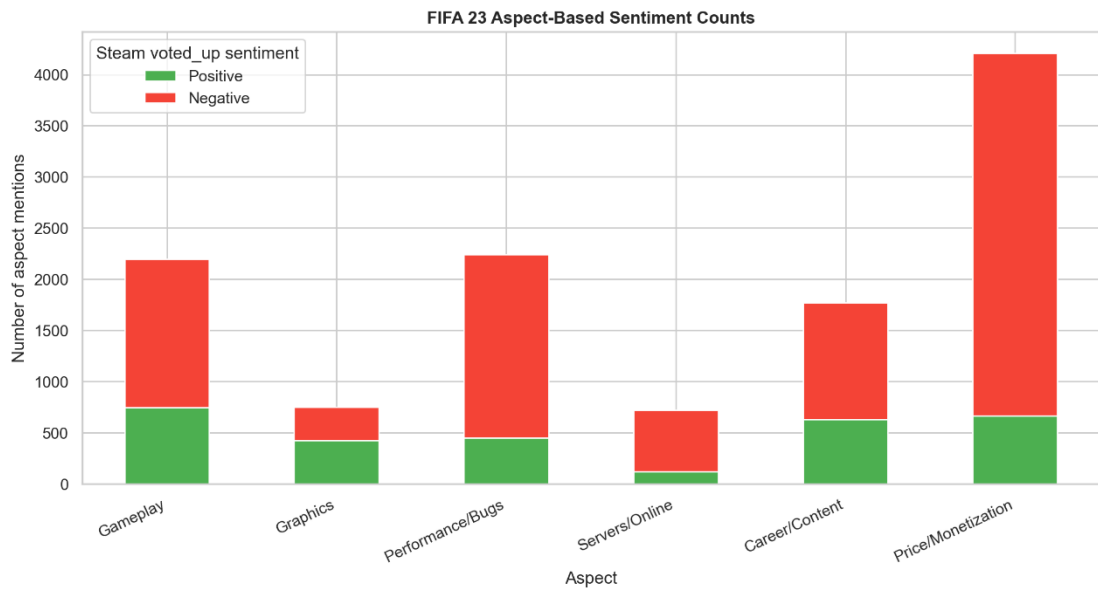


Figure 7.10: Aspect-based positive and negative sentiment distribution for FIFA 23 Steam reviews.

In general, the findings indicate that monetization/value concerns, online server issues, performance, bugs, and anti-cheat complaints were key factors in FIFA 23's negative Steam reception. Some gameplay experiences, popular modes (like Ultimate Team), and graphics did get positive feedback, but were not enough to cure the dissatisfaction in other major areas of the game.

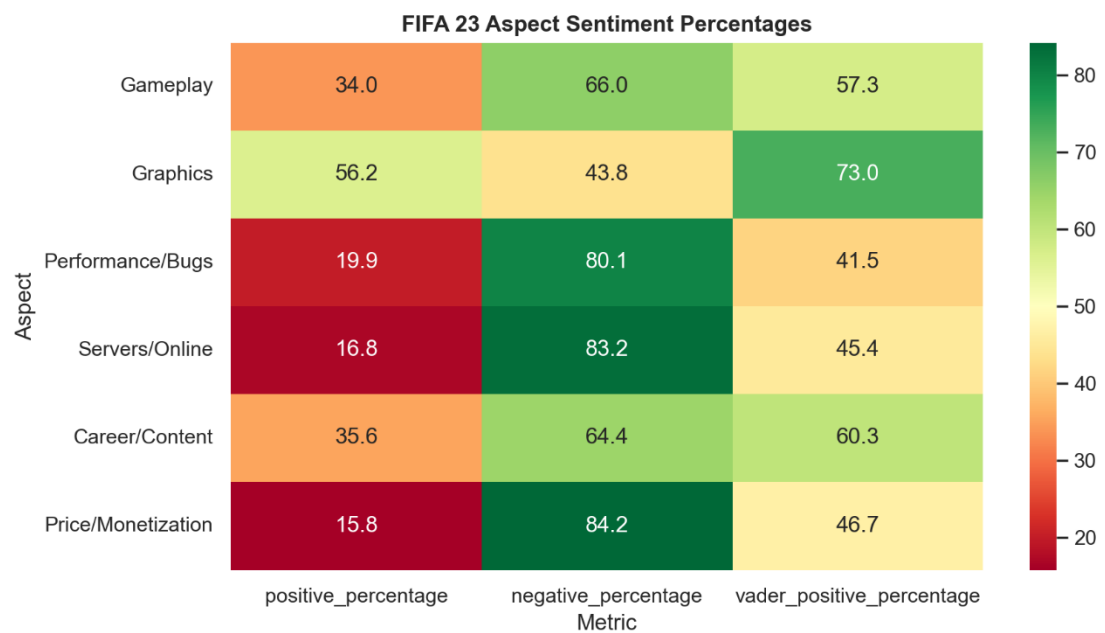


Figure 7.11: Heatmap of positive and negative sentiment percentages by game aspect.

The second multi-class sentiment classification experiment was performed with weak labels from VADER. In this experiment, the votes were not only binary (voted_up=1 or voted_up=0), but the review text was also categorized into three classes: Negative, Neutral, and Positive. The aim is to make a comparison of model behavior in the presence of neutral written-sentiment class. The results however should not be treated as a human annotated sentiment classification since the labels were based on the compound scores from the VADER model.

Table 5 displays the results of the multi-class model comparison. Logistic Regression had the highest overall accuracy (0.8511), macro F1-score (0.8511) and weighted F1-score (0.8503). The second best result was obtained by linear SVM with macro F1 score of 0.8473. The multi-class BiLSTM Neural Network had a score of 0.8450 for macro F1-score very close to the Linear SVM and better than Random Forest and Multinomial Naive Bayes. The above indicates that the BiLSTM model was effective in the multi-class task, but Logistic Regression was the top performing model in general.

Table 7.5: Multi-class sentiment classification results using VADER-derived weak labels.

Model	Accuracy	Macro F1	Weighted F1
Logistic Regression	0.8511	0.8511	0.8503
Linear SVM	0.8474	0.8473	0.8463
BiLSTM Neural Network	0.8429	0.8450	0.8436
Random Forest	0.7890	0.7883	0.7872
Multinomial Naive Bayes	0.6845	0.6630	0.6670

The multi-class results are similar, as the binary classification results were. Logistic Regression and Linear SVM show good results as the TF-IDF features are effective in capturing the words related to sentiment. Another advantage of the BiLSTM model is that it is able to extract patterns from the review text that are sequential. But in the multi-class setting, Multinomial Naive Bayes is less effective, which could be attributed to the difficulty in separating the Neutral class with simplistic word probability specifications.

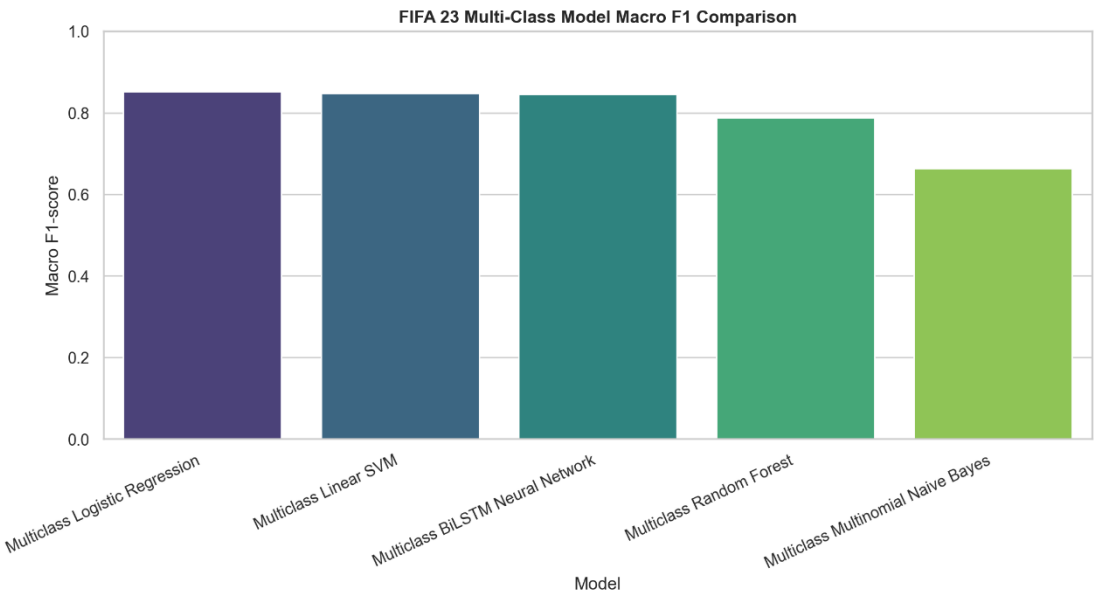


Figure 7.12: Macro F1-score comparison for multi-class sentiment models.

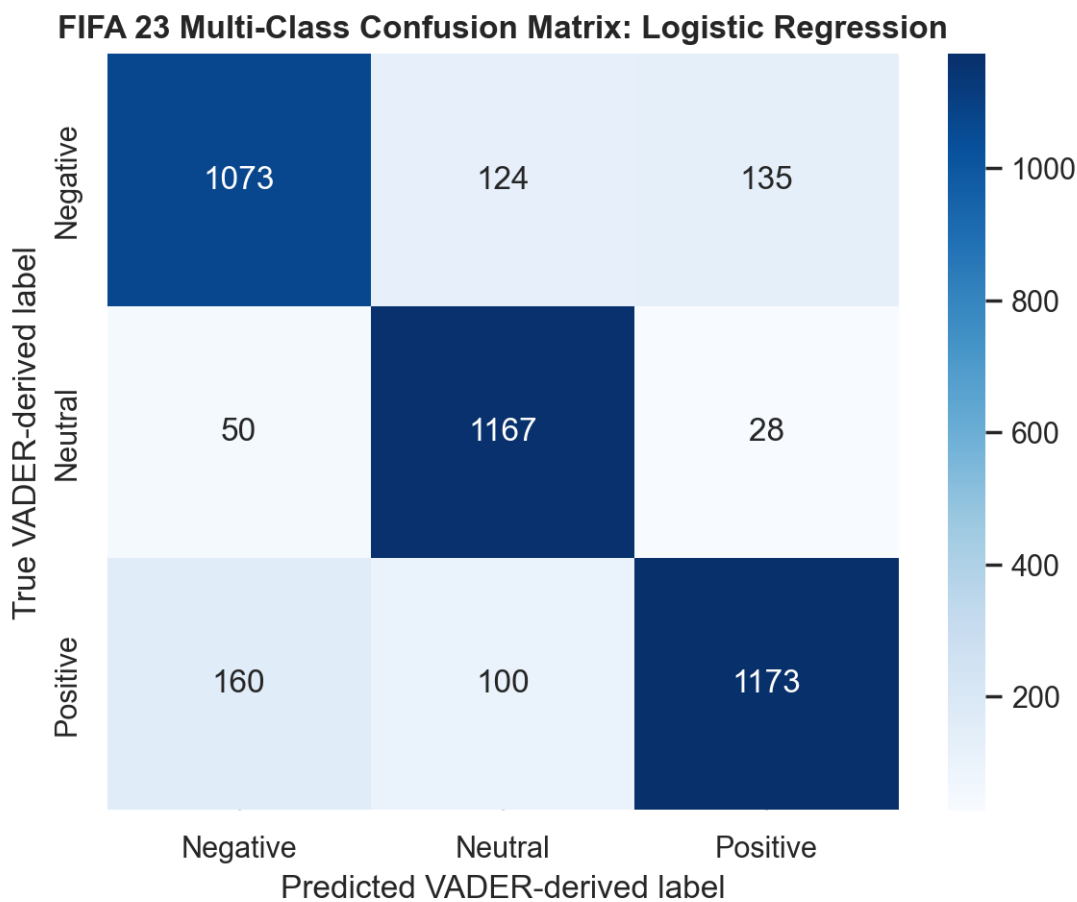


Figure 7.13: Confusion matrix for the best multi-class model (LR).

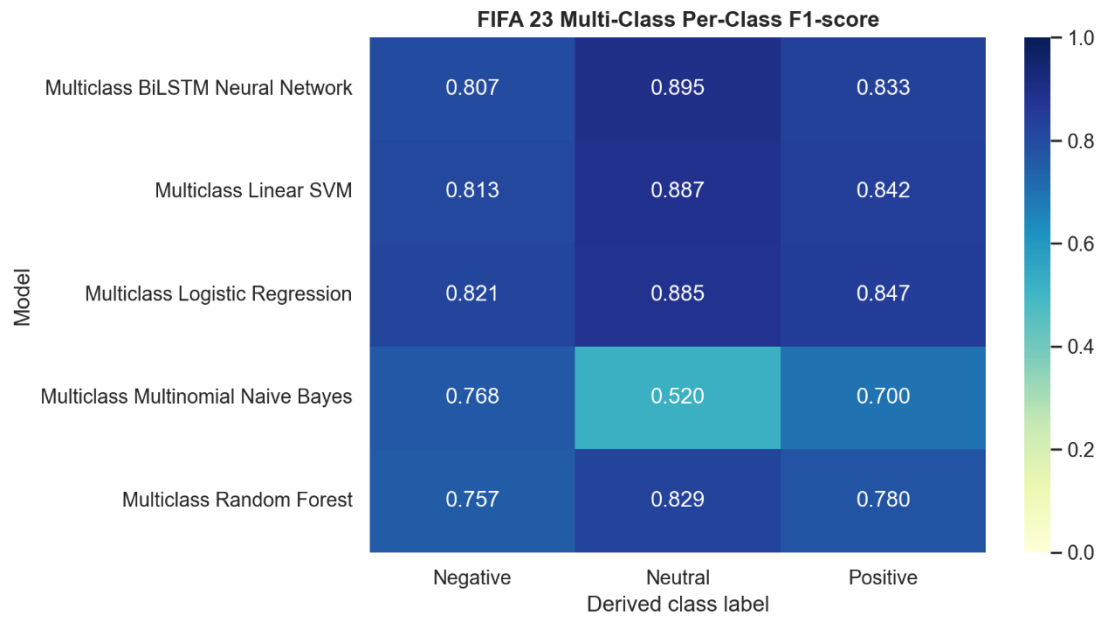


Figure 7.14: Per-class F1-score heatmap for multi-class sentiment models.

8. Discussion

According to the results, Steam users were neither satisfied nor fully satisfied with FIFA 23, but rather they had some negative comments. There is more negative feedback than positive, indicating that many players were frustrated with the game over the period analysed. But the findings also indicate that none single factor was responsible for the negative sentiment. Rather, the dissatisfaction of the players lay in several interconnected aspects, most notably monetization, technical malfunction and bugs, online servers and anti-cheat issues.

Logistic Regression with TF-IDF features was the best performing model. This is a reasonable performance as TF-IDF is used to good effect in short and medium-length review documents where the majority of the most important keywords are strong indicators of sentiment. Additionally, Logistic Regression is also capable of managing high dimensional sparse text features and is much easier to interpret than more complex models. Linear SVM, which also agrees with the suitability of traditional linear classifiers for text classification tasks, came second to the best.

The performance of the BiLSTM Neural Network is found to be good, and the project contains a genuine-trained deep learning model. The F1-Score of 0.8231 indicates the model had a good ability to learn from the cleaned FIFA 23 review text. But it was still slightly outperformed by Logistic Regression and Linear SVM. One reason is that there are lots of short, direct, key-word oriented texts in the review data set. In the case of this kind of data, TF-IDF and linear classifiers can work well, as the words “broken,” “lag,” “good,” “fun,” “refund,” and “crash” are directly relevant to the sentiment and recommendation decisions.

The training history plot for the BiLSTM indicates that overfitting was monitored during training. Training performance improved in epochs and when this improvement was not observed anymore, training was stopped by using validation performance. That's why we added early stopping. While there is a slight difference between training and validation behaviour, as is typical of neural networks, the test F1 score of 0.8231 demonstrates reasonable generalisation. The cross validation results for the old models also are stable, particularly for Logistic Regression and Linear SVM.

In this project the techniques Grid Search and SMOTE are not used. Grid Search was not applied as several different models were already being used in the project, such as

traditional ML, BiLSTM, DistilBERT, VADER, and multi-class experiments. Hyperparameters were set as fixed to allow the comparison to be manageable and reproducible. Additionally, SMOTE was not used since the binary class distribution was not too imbalanced and the multi-class labels derived from VADER were not too imbalanced. In contrast, stratified splitting and F1-score were applied to minimize the evaluation bias.

The BiLSTM model is a sequential learning and word embedding model that can learn the relationship between word order and context better than TF-IDF. Typically, however, neural networks need more data, careful tuning and longer training time to be better than strong traditional baselines. Even though the BiLSTM result was not as good as Random Forest, Multinomial Naive Bayes or the pretrained DistilBERT benchmark was not as good, it is worth noting that the result was better than those of them. This indicates that a small neural network is capable of learning meaningful patterns in the reviews, although Logistic Regression is still the most appropriate model for this data set.

The pretrained DistilBERT SST-2 transformer model was not performing as well as the supervised TF-IDF models or the trained BiLSTM model. This finding is significant because it demonstrates that a higher-level, pre-trained model is not necessarily superior, when it is not fine-tuned with the same label structure and on the same domain. It is a recommendation label, whereas DistilBERT is able to grasp both positive and negative language. Even when the review is a mixed bag, the players can either recommend or not recommend the game for overall value, technical issues, online stability, anti-cheat issues or their expectations. The DistilBERT result shows the necessity of adapting transformer models to the specific task with task-specific data. The results from the hypothesis testing offer valuable insights into player behaviour. Positive reviewers spent much more time playing recently than the negative reviewers. This indicates that players who were still engaged in the game of FIFA 23 were more likely to recommend it. The median number of games owned, on the other hand, was higher among negative reviewers. This could signal that a player who has more gaming experience will be able to make a comparison with the other games and therefore may make a more critical judgment about FIFA 23.

The sentiment results for each month reveal a shift in player sentiment throughout the month. Sentiment can change after launch, after updates, when there are server issues, patches, or changes to the gameplay in gaming communities, as this is to be expected. Sentiment is not set at release for live-service and online. It can either improve or deteriorate based on the rate of which the developers respond to player grievances.

It can also be seen from the VADER comparison that the written sentiment and Steam recommendation are correlated but not the same. The agreement rate is 69.36%, which means that VADER successfully identifies a lot of cases, but does not catch them all. Gaming reviews often include sarcasm, slang, short reviews, mixed opinions, etc. For instance, they can write nice things about anti-cheat when they complain about it, or they can write negative but emotional comments on the losing side of an online game. This is why it is not sufficient to use lexicon based sentiment analysis on this data set.

The distribution also differs for the three classes of VADER, which is why we took the binary classification approach to have to be careful in this project. VADER is able to classify written text as positive, neutral, or negative, and the descriptive result indicates that there is a significant group of neutral text. The supervised data set, however, doesn't have a neutral label on the recommendation. One problem with the Steam data is that there is no way to annotate it without adding an extra neutral class, because Steam only supports `voted_up=True` and `False`, not a neutral class. Therefore, the primary classification task is binary, and the VADER 3-class is only provided as a description of the sentiment tone in the written text.

The multi-class experiment includes a Neutral class, giving another perspective on written sentiment. However, this experiment is different from the main binary classification task. The binary models are trained with the real Steam `voted_up` label and the multi-class models are trained with the weak labels derived from VADER. Therefore, the multi-class results represent the ability of the models to predict the polarity classes as written by humans rather than the ability to predict the manually annotated human emotion.

In the multi-class experiment, Logistic Regression had the highest macro F1-score. This indicates that the TF-IDF features are still very powerful for sentiment classification in reviews even if a Neutral class is included. The BiLSTM model also achieved a good performance and the macro F1 score was 0.8450, indicating that a neural network can

successfully learn useful three-class sentiment patterns from the text of the reviews. But, as the multi-class labels are generated from VADER, any limitations of VADER (such as inability to handle sarcasm, slang, gaming terms, or multiple opinions) can also affect the multi-class model results.

Care needs to be taken with interpretation of the Neutral class as well. In this project, Neutral does not mean that a human annotator judged the review to be neutral. The VADER compound score is in the range of -0.05 to 0.05. The reviews could be brief, descriptive, ambiguous, or devoid of emotion. The multi-class experiment provided a meaningful comparison and analysis, but the binary `voted_up` task is the primary supervised classification task.

The aspect-based sentiment results are most useful for making practical recommendations. Price/Monetization, Servers/Online, and Performance/Bugs are the most negative aspects. These are the first areas that should be targeted by developers as they influence whether or not players feel they're playing a fair, stable and worthwhile game. The most positive aspect was graphics, indicating that the main reason for dissatisfaction was not due to graphics. So the improvements of technical stability, online reliability, anti-cheat systems, and value perception would likely have a greater influence on player satisfaction than any improvements to visuals alone.

This project also has some restrictions. Firstly, the `voted_up` label is considered as the supervised sentiment label, however it is not a measure of emotion but rather a measure of recommendation behaviour. Second, pre-defined sentiment keywords are required for the aspect-based sentiment analysis, which can cause it to fail to recognize opinions that are expressed with alternative words or expressions. Third, the transformer model was employed without fine-tuning with the FIFA 23 dataset. The project offers a valuable and informative snapshot of player sentiment towards FIFA 23 and the key factors contributing to positive and negative Steam reviews, providing valuable insights despite these constraints.

9. Conclusion & Future Work

The aim of this project was to create an end-to-end system for sentiment analysis and opinion mining of FIFA 23 Steam reviews. The Steam voted up column was used as the supervised labels in the study, with positive reviews as the label for recommended reviews and the negative reviews as the label for not recommended reviews. The data set was cleaned before analysis, resulting in 20,050 reviews being analysed, of which 9,310 were positive and 10,740 were negative.

This shows that FIFA 23 has mixed reviews on Steam with negative ones being slightly more than positive ones. Logistic Regression had the highest overall model performance with an F1 score of 0.8468. The second best model was Linear SVM and the trained BiLSTM Neural Network had an F1-score of 0.8231. To meet the criteria of deep learning, the BiLSTM model was used and trained directly on the cleaned FIFA 23 Steam reviews by applying text vectorization, embeddings, Bidirectional LSTM layer, dropout, and early stopping. It is not able to perform better than Logistic Regression but better than Random Forest, Multinomial Naive Bayes and pretrained DistilBERT model for SST-2 benchmark.

Hypothesis testing results revealed that there was a significant relationship between sentiment and player behaviour. Higher-playtime players were more likely to leave positive reviews, and more game-owning players were more likely to leave negative reviews. Sentiment also evolved heavily over time and indicates that player opinion is not stable over the review period. The 3-class Positive, Neutral, and Negative distribution of VADER sentiment was first explored, but was modified to a binary sentiment to make direct comparisons with the Steam voted_up label. This is required since there is no neutral class in the voted_up class. The binary VADER sentiment showed a close but non-exact match to Steam voted_up labels on cleaned reviews (69.36%). This indicates that there is a relationship between written sentiment and voted_up labels, but it is not exactly the same.

The opinion mining and aspect based sentiment analysis gave a deeper understanding of why the players gave feedback. The most common phrases in positive reviews were “good game”, “best game” and “great game”, and the most common phrases in negative reviews were “anti cheat”, “worst game”, “bad game”, “fix game” and “cant play”. The most negative aspects were Price/Monetization, Servers/Online, and Performance/Bugs.

The most positive was "Graphics". They indicate that player dissatisfaction had more to do with perceived value, online stability, technical issues and issues of the gameplay itself, rather than "visual quality".

There were a number of difficulties encountered in the project. A problem presented itself in cleaning up the messy text from reviews, as Steam reviews feature informal language, repetition, slang, and short comments. Another difficulty was understanding the `voted_up` label, which refers to the status of a recommendation, not just to the feeling. The transformer model also found that there are some differences between general pretrained sentiment models and domain-specific recommendation labels when they are not fine-tuned. Both raw and cleaned review text was stored, supervised labels were compared to written sentiment from VADER, and error analysis techniques were used to develop an understanding of mismatches.

There was also an experiment with multi-class sentiment classification using weak labels derived from VADER. In this experiment, the reviews were categorized as Negative, Neutral, and Positive according to the VADER thresholds of the compound score. The overall performance of multiple classes was best for Logistic Regression with a macro F1 score of 0.8511, followed by Linear SVM and the BiLSTM Neural Network. These results indicate that the models are able to learn the written sentiment categories effectively, like those of VADER. The multi-class labels are not human annotations, but the results generated should be regarded as weak label sentiment classification rather than human labelled emotion detection.

In future, a manually annotated sentiment data set could be developed by manually labelling a subset of FIFA 23 reviews as positive, neutral, or negative. This would allow a more reliable multi-class classification task instead of relying on VADER-derived weak labels. Future research could also fine-tune DistilBERT or another transformer model directly on FIFA 23 Steam reviews, particularly identifying neutral, mixed, sarcastic or context dependent gaming reviews. Hyperparameter tuning, larger embedding dimension, sequence length, utilizing pretrained word embedding, as well as comparing the BiLSTM model with GRU and CNN-based neural networks, could be used to enhance the model. Football-specific and EA-specific aspect terms can also be added to the aspect dictionaries to enhance the aspect-based sentiment analysis.

In conclusion, this project demonstrates the potential of sentiment analysis and opinion mining to gain valuable insights into user experiences in online gaming forums. The results can guide game developers and publishers in grasping player preferences, areas of dissatisfaction, and areas for improvement.

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Appendix A

The GitHub Repository for this project contains the main notebook and the FIFA 23 Steam Reviews dataset used in the analysis and some other markdown files for explanation purposes.

Repository link on GitHub:

<https://github.com/Yaserabulaban/SMC-Assignment>

NOTE: you will need to request access to the repository since it is private

Google Drive folder link:

<https://drive.google.com/drive/folders/1Dda1o18SCIfsVS6hnYdRTd5BIZ8TYx0p?usp=sharing>

The Google Drive folder contains the Google Colab notebook, the FIFA 23 Steam reviews dataset, and related output files used for this project.

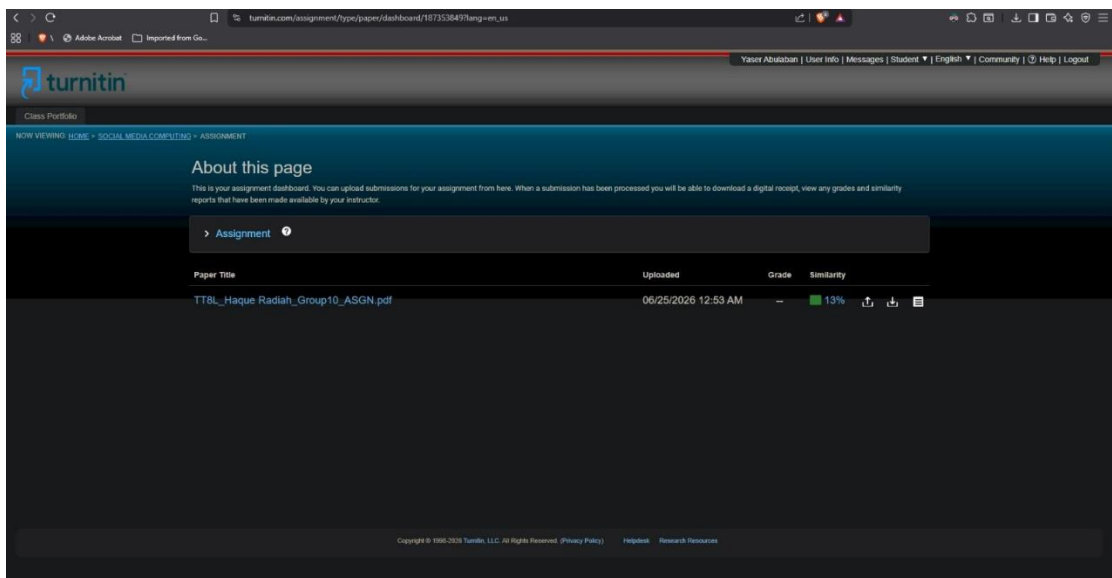
Included files:

- tutorial_1_smc.ipynb
- fifa23_steam_reviews.csv

The notebook contains the full implementation of data cleaning, preprocessing, exploratory data analysis, VADER sentiment analysis, binary supervised classification, BiLSTM deep learning, DistilBERT transformer benchmark, multi-class VADER-derived sentiment classification, opinion mining, aspect-based sentiment analysis, and result visualization.

Appendix B

Turnitin Similarity Percentage



The screenshot shows a Turnitin assignment dashboard in a web browser. The URL is turnitin.com/assignment/7type/paper/dashboard/187353843?lang=en_us. The user is Yaser Abuabbar, and the page is titled "About this page". Below the title, there is a section for "Assignment" with a dropdown menu. A table lists the submission details:

Paper Title	Uploaded	Grade	Similarity
TTBL_Haque Radiah_Group10_ASGN.pdf	06/25/2020 12:53 AM	--	13%

The similarity score is 13%, indicated by a green bar. At the bottom of the page, there is a copyright notice: "Copyright © 1998-2020 Turnitin, LLC. All Rights Reserved. (Privacy Policy) | Helpdesk | Research Resources".